# Chapter 1

# INTRODUCTION

This chapter introduces the study, laying the groundwork for research by presenting its fundamental components. It begins by discussing the problem and its background to establish the context and rationale for the research. The chapter then reviews related literature and studies to situate the research within the existing body of knowledge and identify gaps. It outlines the theoretical and conceptual frameworks guiding the study, followed by a clear statement of the problem and the corresponding hypothesis or assumptions. The chapter concludes by explaining the significance of the study, its scope and limitations, and a definition of terms used throughout the paper.

## The Problem and Its Background

Academic dishonesty continues to be a serious issue at City of Malabon University (CMU). This is not limited to the College of Computer Studies (CCS), as the issue also occurs in all other departments in the university. As researchers observed and heard reports from students and professors about multiple instances of cheating, including the usage of gadgets, such as smartphones, during tests, cheat sheets, and leaked exams. What is occurring in other Philippine colleges is reflected in these issues. For example, cheating services have become more common among college students, often because of pressure to perform and easier access to digital tools (Aguilar, 2021).

During the pandemic, when academic institutions switched to online learning platforms, many teachers in the Philippines noticed that cheating became more prevalent. A recent study showed that a majority of teachers saw more dishonest behavior during remote exams, mainly because of the lack of proper monitoring (International Journal of Learning, Teaching and Educational Research [IJLTER], 2023). These findings align with students' experience at CMU, where some students view cheating as low-risk when monitoring is weak and penalties are unclear.

In a local context, studies found that 9 out of 10 students admitted to cheating at least once over an academic year and that 8 out of 10 students in Cagayan had cheated on exams, underscoring the need for stronger academic integrity measures (Quintos, 2017; De La Cruz & Bautista, 2023). Both studies confirm that cheating is prevalent in Philippine colleges.

One major factor is the limitation of human proctoring. At CMU, one proctor is usually assigned to about 40 students. In this kind of setup, it is very hard to watch the students closely. This makes it easier for some students to cheat without being caught. Similar problems are also reported in other schools, where teachers mention issues like blocked views and not enough time to monitor everyone properly (IJLTER, 2023).

To help solve this problem, the researchers propose to develop a program called Cheat GPT. A system that will use live camera footage and computer vision to detect common signs of cheating, such as repeated looking to the side,

suspicious hand movements, or the use of gadgets. The tool sends real-time notifications to help proctors respond immediately. The researchers believe this tool can help improve honesty during exams and support fair grading in CMU and other schools in the Philippines.

The following sections will elaborate on the technical architecture, implementation strategy, ethical safeguards, and broader implications of deploying AI-based integrity tools, such as Cheat GPT, in Philippine universities.

## Review Related Literature and Studies

### *Gesture Recognition Techniques in Real-Time Video Stream*

Gesture recognition in real-time video streams is foundational for systems that aim to understand human behavior through visual data. In the context of academic exam monitoring, effective gesture recognition allows AI systems to detect suspicious behaviors without relying on invasive methods. This theme integrates research that spans cheating detection, pose estimation, real-time video processing, and multi-person gesture tracking. An amalgamation of challenges across the studies on how to balance accuracy, speed, and scalability, especially in complex, crowded environments like classrooms.

Academic cheating has been explored through technological approaches such as machine learning and pose estimation. One study designed a machine learning system that detects student postures indicative of dishonesty, utilizing OpenPose and Extreme Gradient Boosting (XGBoost) classifiers to achieve 90%

accuracy and real-time performance (~10 frames per second), which was positively reviewed by human proctors in classroom simulations (Bancud, 2021).

A deep learning-based system for monitoring students' attention and behavior in real-time classroom settings was also proposed, utilizing You Only Look Once (YOLOv5) for object and gesture detection and DeepSORT for behavior tracking. By identifying gestures such as "raising hands" (close attention) and "using phones" (low attention), the system successfully linked visual behavior to attention states, achieving a Mean Average Precision (mAP) @0.5 of 0.877 for emotion detection and 76% accuracy in classifying behaviors. In contrast to the earlier work that emphasized posture for cheating detection, this study expanded the behavioral context to include engagement and emotion (Trabelsi et al., 2023).

While earlier works applied pose estimation to specific contexts, broader frameworks like OpenPose and AlphaPose provide the technical backbone. OpenPose introduced Part Affinity Fields (PAFs) for fast, multi-person detection of body, face, hand, and foot keypoints (Cao et al., 2021). In contrast, AlphaPose enhanced both precision and identity tracking with Symmetric Integral Keypoint Regression and pose-aware re-ID features (Fang et al., 2022). Together, these models support scalable, real-time gesture tracking and are widely used as base architectures in modern surveillance and monitoring systems.

Several researchers also addressed gesture tracking challenges in crowded scenes, which are common in classroom environments. One approach introduced a Keypoint Confidence Network to handle occlusions and missing

identities, achieving strong Multiple Object Tracking Accuracy (MOTA) scores on PoseTrack datasets (Fu et al., 2022). Another developed DetPoseNet to filter noisy detections and refine keypoints, showing high speed and robustness using coarse-pose filtering (Ke et al., 2021). Both studies support the need for occlusion-aware gesture detection in real-time, multi-person scenes like exam halls.

A review of the broader hand gesture recognition field identified computational constraints, user adaptability, and gesture ambiguity as ongoing limitations, supporting earlier findings that emphasized the need for early gesture detection and filtering (Mohamed et al., 2024).

Finally, gesture analysis was extended into affective computing through a system that used facial Action Units, eye openness, and head pose to detect depressive states in real-world settings. While the context differs, the successful integration of head movement and eye gaze in real-time from mobile devices aligns with this capstone's interest in subtle cheating cues such as frequent head turns or gaze aversion (Islam & Bae, 2024).

***Machine Learning for Detecting Unauthorized Device Use in Classroom Exams***

Detecting unauthorized device use during exams has become a focal point for machine learning researchers aiming to develop automated, real-time monitoring systems. One innovative approach enhanced the You Only Look Once (YOLOv8) object detection model by integrating an Efficient Channel Attention (ECA) mechanism, significantly improving the system's sensitivity to subtle

behaviors such as discreet phone use. This model was trained on a large dataset derived from classroom footage and demonstrated superior accuracy over traditional methods like Single Shot MultiBox Detector (SSD) and Faster Region-based Convolutional Neural Network (Faster R-CNN). Running at 27 frames per second, it was considered practical for real-time exam surveillance (Zuo et al., 2024).

Beyond classroom-specific studies, applications in industrial environments have also provided useful insights. A system developed for detecting unauthorized phone use in restricted zones utilized You Only Look Once (YOLOv8) combined with a Residual Network-50 classifier, achieving impressive precision while maintaining fast inference speeds. Although originally intended for workplace safety, the challenges it faced with occluded devices and varying lighting highlight common difficulties that also arise in exam proctoring contexts (Deshpande et al., 2025).

An intelligent exam monitoring system using Faster Region-based Convolutional Neural Network for motion detection and Multi-task Cascaded Convolutional Networks (MTCNN) for face recognition was developed to detect head orientation and potential cheating behaviors. Though the main focus was on head orientation, the authors noted that a Faster Region-based Convolutional Neural Network could be extended to detect objects like mobile phones. The system achieved 98.5% accuracy in identifying head-turning behaviors and 95%

in facial recognition, making it a strong candidate for integration with phone detection modules (Mahmood et al., 2022).

A contextual action recognition system using a Support Vector Machine (SVM) classifier was created and tested on five cheating behaviors, including phone usage, on a custom dataset. Speeded Up Robust Feature (SURF) feature descriptors outperformed others, achieving 90% accuracy in detecting cellular devices. However, the system struggled with occlusion and object blending (e.g., dark phones against dark clothes), highlighting real-world complexity (Hussein et al., 2022).

A You Only Look Once (YOLOv5) based surveillance system that monitored multiple classrooms in parallel was implemented, achieving 85.7% accuracy in detecting phone use and optimized for scalability with minimal hardware requirements (Jayakumar & Peddakrishna, 2024). A Transformer-based attention model to detect concealed device use was also proposed, achieving 93.4% accuracy and proving effective at recognizing subtle cues like hiding a phone under the desk or inside clothing (Pereira & Hussain, 2024). Researchers focusing directly on exam environments have fine-tuned advanced object detectors like Faster Region-based Convolutional Neural Network to recognize phones with remarkable accuracy. For instance, a study using a Residual Network 50 backbone on a dataset from simulated exam scenarios achieved near-perfect accuracy but still struggled to detect partially hidden phones. The authors

emphasized the value of increasing dataset size and incorporating multi-camera views to reduce false negatives and improve robustness (Moyo et al., 2023).

A real-time smartphone detection system using the YOLOv5 algorithm was developed to monitor unauthorized device use in classrooms. Trained on a dataset of 1,038 images, the model achieved 77.7% mean average precision (mAP), 93.2% precision, and 71.7% recall. The system successfully detected smartphones under various lighting conditions and orientations, but lacked automatic notifications, requiring manual monitoring. This research demonstrates the potential of machine learning for real-time surveillance to prevent distractions and cheating in classroom environments (Sejati et al., 2023). Similarly, a multimodal monitoring system combining OpenCV's Haar Cascades for object detection with microphone-based audio surveillance identified phone usage and whispering in real time, demonstrating how simple tools can achieve meaningful results in practical deployments (Radhika et al., 2023).

Incorporating multiple detection strategies, the Copying Mechanism in the Sequence-to-Sequence Learning framework merges the strengths of You Only Look Once (YOLOv5) and Faster Region-based Convolutional Neural Network with deep feature extraction and temporal analysis via optical flow. This hybrid model excels not only at detecting phones but also at identifying suspicious behavioral patterns in face-to-face exams without requiring intrusive biometric data. Its design reflects a growing concern for privacy-conscious AI solutions in

educational settings, balancing detection accuracy with ethical considerations (Sirt & Saykol, 2023).

***Ethical Implications of Surveillance and AI-Powered Monitoring Systems***

As educational institutions adopt AI-powered surveillance tools to monitor exam behavior, ethical concerns related to privacy, consent, data protection, and algorithmic fairness emerge. This theme explores literature on the ethical dimensions of surveillance technologies, particularly as applied to real-time monitoring systems. While studies span sectors such as policing, healthcare, and smart cities, their findings are directly applicable to classroom-based behavioral monitoring systems like Cheat GPT.

A real-time attention monitoring system for classrooms using deep learning was developed, obtaining informed consent, minimizing data retention, and following data protection laws. Still, researchers noted risks of behavioral bias, mislabeling, and student discomfort, particularly if real-time feedback is used punitively. This study aligns with another that proposed a privacy-preserving surveillance model using pose-based tracking, demonstrating that anonymized systems can comply with privacy laws such as the General Data Protection Regulation (GDPR) while maintaining functionality. However, it also acknowledged a performance trade-off, with reduced anomaly detection accuracy, supporting the need to balance privacy and utility in educational surveillance tools (Trabelsi et al., 2023; Ardabili et al., 2023).

A comparative legal analysis of facial recognition in the US, EU, and UK found that educational institutions often lack proper oversight. The study highlighted a case in Sweden where a school was fined for using Facial recognition technology without consent. The findings suggest that regulatory gaps, consent imbalances, and algorithmic bias continue to threaten students' rights when biometric systems are used in classrooms (Almeida et al., 2022).

Research investigating student awareness of AI's ethical implications in Zamboanga City, Philippines, found that 55% of respondents were concerned about data privacy and job displacement. Focus group participants voiced mistrust in AI systems due to a lack of transparency and consent, echoing global concerns but rooted in a local educational context (Valerio, 2024). Similarly, another study explored the ethical implications of AI in schools, focusing on data privacy, surveillance, and algorithmic bias. The literature review revealed that AI systems often collect extensive personal data without meaningful consent, and tools such as emotion tracking contribute to increased student anxiety. The authors emphasized the psychological toll of constant monitoring and the structural inequalities these systems can reinforce (Shukla et al., 2025).

In higher education, AI integration raises significant ethical issues, especially when used for assessment, engagement tracking, and proctoring. Studies found that algorithmic bias and data misuse are common, and many institutions lack both regulatory policies and ethics education to guide implementation. This finding supports previous research stating that the

educational sector is underprepared for the ethical challenges posed by AI surveillance (Hutson et al., 2022; Shukla et al., 2025).

AI surveillance across sectors, especially in education, frequently lacks transparency and informed consent. A bibliometric review of 279 publications revealed that data collection practices often occur without user knowledge, and marginalized groups are disproportionately affected by algorithmic decisions. This supports prior work highlighting the dangers of opaque AI systems in education (Debnath et al., 2024; Valerio, 2024).

AI surveillance shares ethical concerns across domains, including public health and education. Key issues include loss of autonomy, biased data interpretation, and insufficient regulation. While system-wide analyses focused on technical and procedural gaps, this perspective added a social science lens, arguing that surveillance disproportionately targets vulnerable populations and erodes trust in institutional settings (Saheb, 2022; Debnath et al., 2024).

Smart policing technologies such as facial recognition and predictive analytics present serious ethical challenges, particularly around transparency, bias, and consent. Qualitative analysis of case studies found that these technologies often function with limited oversight, disproportionately affect marginalized communities, and risk mission creep expanding beyond their original purposes. Although focused on law enforcement, these findings raise critical warnings for educational surveillance, where similar dynamics may occur (Chris et al., 2025).

Perceived usefulness (PU) of AI surveillance often overrides ethical concerns in user decision-making. Findings revealed that individuals are more likely to accept surveillance when they believe it enhances outcomes—even in ethically questionable scenarios. In educational settings, this suggests that students and teachers may accept systems like Cheat GPT if framed as necessary for exam integrity, even without full ethical transparency (Anton et al., 2021).

***The Role of Emerging Technologies in Modern Cheating Detection Systems***

Emerging technologies such as AI, biometrics, and surveillance-based monitoring tools have increasingly become central to academic integrity enforcement. While these innovations aim to improve cheating detection, their implementation reveals significant ethical, social, and cultural complexities. A recurring pattern across the literature is the tension between efficiency and invasiveness. While AI and biometric tools promise scalable oversight, they often compromise transparency, student agency, and institutional control. Common themes include privacy concerns, bias in algorithmic decisions, and resistance from students and educators alike. Differences arise in how technologies are deployed across institutions, as well as in the ethical and policy frameworks that attempt to govern them.

Emerging technologies such as facial recognition, keystroke tracking, and behavioral analytics are redefining how academic dishonesty is monitored in higher education. This qualitative study critically analyzes AI and biometric-based proctoring systems through a case-based and literature review approach, focusing

on the United States and the United Kingdom. The study found that AI-powered systems increasingly calculate “integrity scores” based on behavior, often without user transparency (Oravec, 2022).

A highly efficient real-time cheating detection system was developed using a combination of deep learning and statistical analysis. The system applied convolutional neural networks (CNNs) to analyze video data and discrete Fourier transform (DFT) techniques for speech detection, achieving detection rates near perfection: 99.83% for the front camera, 98.78% for the back camera, and 100% for speech. The researchers suggested that such systems could revolutionize online exam administration, reducing the burden on human proctors (Kaddoura & Gumaei, 2022).

Research explored how obfuscation techniques such as deepfakes, silhouette replacements, and 3D avatars can serve as privacy-preserving tools in remote proctoring systems without significantly compromising cheating detection capabilities. Using a mixed-methods design, the study combined interviews with nine domain experts and a vignette-based experiment involving 259 UK-based participants. Results showed that 3D avatars and deepfake-based body substitutions offered the best balance between maintaining information fidelity and respecting privacy and fairness (Mukherjee et al., 2024).

The effectiveness of deep learning and machine learning algorithms in detecting cheating behaviors during online exams was examined. A five-layer deep neural network (DNN) achieved 80.9% accuracy in regression-based detection. In

binary classification, a ten-layer DNN model reached 96.9% accuracy, while the Extreme Gradient Boosting (XGBoost) model outperformed others in triple classification tasks with 97.7% accuracy. These findings underscore the potential for scalable, data-driven systems to uphold academic integrity and reduce reliance on manual supervision in distance learning environments (Erdem & Karabatak, 2025).

A multi-perspective, adaptive cheating detection system tailored for paperless online examinations was designed. The system integrates three distinct camera angles: overhead, facial, and horizontal, and uses a Swin Transformer for gaze tracking, Lightweight- You Only Look Once (YOLOv5) for cheating tool detection, and a multilayer perceptron (MLP) for behavior classification. The system achieved a cheating behavior detection accuracy of 95.4% at 35 FPS, with object detection accuracies exceeding 97%. According to the authors, this adaptive system minimized blind spots in surveillance, reduced student misconduct, and significantly lowered the need for human invigilators (Hu et al., 2024).

A comprehensive literature review and case study analysis were conducted to explore the breadth of cheating detection and prevention technologies in university-level online examinations. The review highlights how AI-based proctoring systems leverage webcam feeds, audio inputs, and gaze or motion analysis to detect dishonesty (Kumar & Rohil, 2024).

A systematic literature review was conducted to examine how machine learning (ML) models contribute to online learning and examination systems, particularly during lockdown scenarios. The findings underscore ML's pivotal role in cheating detection through tools that incorporate computer vision, behavioral analytics (e.g., eye tracking, posture recognition), and biometric authentication using facial, iris, and keystroke dynamics. Proctoring platforms such as ProctorU and Respondus were identified as prime examples of ML-enabled systems that automate supervision by monitoring screen activity, sound, and facial expressions in real time. The study also highlights the application of Machine Learning for plagiarism and anomaly detection using algorithms such as Levenshtein distance, cosine similarity, and long short-term memory (LSTM) networks. Additionally, hybrid authentication models combining multiple biometric features were found to be effective in reducing impersonation risks. These insights reiterate that while Machine Learning based systems enhance exam integrity, their ethical and infrastructural implications must be addressed to ensure equitable deployment (Kaddoura et al., 2022).

The ethical dimensions of AI-powered proctoring systems used in online exams were explored. While these systems utilize facial recognition, head movement tracking, and environmental scanning to detect misconduct, the authors argued that such practices often produce false positives, lack transparency, and risk reinforcing biases against neurodiverse students and those from minority groups. The systems, often deployed without adequate consent, are seen to

violate student privacy by monitoring personal spaces like bedrooms. The authors advocate for ethical governance, transparent design, and the offering of less invasive alternatives to ensure fairness while still protecting academic integrity (Coghlan et al., 2020).

Generative AI tools like ChatGPT and Bard complicate the academic cheating landscape. Through theoretical and literature-based analysis, research revealed that while generative AI can facilitate dishonesty, it also offers tools for detection, such as watermarking, version tracking, and perplexity analysis. The study emphasized transparency and ethical safeguards over punishment and called for revised academic integrity policies to address this evolving frontier. It adds depth to the conversation on the dual role of emerging technologies in both enabling and mitigating academic misconduct (Oravec, 2023).

A systematic literature review was conducted to investigate the evolution and implementation of digital proctoring in higher education. Using Latent Dirichlet Allocation (LDA) topic modelling, the study identified key themes related to cheating detection and academic integrity. The findings highlighted the integration of emerging technologies in two areas: AI-driven surveillance systems for real-time behavior analysis and biometric authentication methods for identity verification. These technologies significantly enhance exam security and uphold academic fairness in online settings. However, the study also documented persistent challenges, particularly privacy concerns, algorithmic bias, and ethical discomfort related to continuous monitoring. The authors recommend multidisciplinary

research and ethical guidelines to ensure future systems are both effective and equitable (Han et al., 2024).

**Synthesis**

The literature reveals a dynamic intersection between artificial intelligence, educational integrity, and ethical governance, segmented across four major themes relevant to classroom cheating detection. These themes together provide a conceptual and technical scaffold for the development of the Cheat GPT system. A clear pattern emerges in gesture recognition techniques in real-time video streams, where studies converge on the viability of using pose estimation for behavior analysis in academic contexts. Studies have demonstrated effective multi-person pose tracking in real-time, a capability that is foundational for detecting body language associated with cheating (Cao et al., 2021; Fang et al., 2022; Ke et al., 2021). Some researchers have adapted these models specifically for classroom surveillance (Bancud, 2021), while others have introduced robustness under occlusion, an important consideration in group exam settings (Fu et al., 2022).

In machine learning for detecting unauthorized device use, studies consistently support the use of deep learning models for identifying mobile phone usage (Zuo et al., 2024; Deshpande et al., 2025; Moyo et al., 2021). These models proved highly effective in object detection tasks, achieving accuracies above 90% even under real-time constraints. They are complemented by posture and head pose models that offer non-invasive alternatives for inferring suspicious behavior

(Kumar et al., 2022; Becerra et al., 2023). However, a point of divergence lies in the level of computational resources required; some systems demand high-end GPUs, which limits their feasibility in low-resource educational institutions.

The tension between surveillance efficiency and ethical responsibility is prominent in discussions surrounding AI-powered monitoring systems. Scholars critique the growing use of AI surveillance for its potential to normalize invasive data collection, especially without meaningful consent (Almeida et al., 2021; Valerio, 2024; Shukla et al., 2025). A trend across these studies is concern over algorithmic bias and the disproportionate impact on marginalized students. Furthermore, researchers emphasize that user acceptance often hinges more on perceived usefulness than ethical clarity, suggesting a disconnect between deployment and understanding (Anton et al., 2021; Debnath et al., 2024).

The role of emerging technology in modern cheating detection reveals a growing interest in multimodal systems. Researchers are beginning to combine gesture, facial, and audio to build more holistic detection tools. For instance, studies have fused audio and gesture data for real-time inference and applied transformer models to detect academic dishonesty (Radhika et al., 2023; Sirt & Saykol, 2023). These methods mark a trend toward more context-aware, flexible, and integrated systems, but the scalability and data privacy of such approaches remain under-addressed.

A critical gap shared across all themes is the lack of a unified system that merges gesture recognition, device detection, and ethical governance within a

lightweight, real-time framework. Most current systems excel in one domain but fail to address others holistically. This fragmentation underscores the rationale behind Cheat GPT, a system designed to combine multi-angle gesture monitoring with real-time inference while maintaining transparency and ethical integrity.

In summary, while there is strong agreement on the technical potential of AI in classroom surveillance, there is less consensus on how to do so ethically, affordably, and at scale. The challenge, therefore, is not only to build accurate systems but to do so with attention to fairness, consent, and user empowerment.

## Theoretical Framework

### *Academic Integrity Models*

The system is based on educational theories that explain why students cheat and how schools can discourage it.

### *Deterrence Theory*

(Becker, 1968) explained that people are less likely to break rules if they believe there is a high chance of getting caught. Our system supports this idea by offering continuous monitoring and real-time detection, which can raise students' awareness of the risks and reduce the chances of cheating. This is relevant to the local setting. In a study on academic dishonesty in the Philippines, many students admitted to cheating because they believed the risk of being caught was low (Beruin, 2022).

***Proctoring Equity Framework***

Fair monitoring during exams is important (Alessio et al., 2017). In some classrooms, especially those with large or uneven seating arrangements, students in certain spots may not be watched as closely. Our system uses camera-based tracking to reduce this bias, helping ensure that all students are monitored equally.

***Practical Integration***

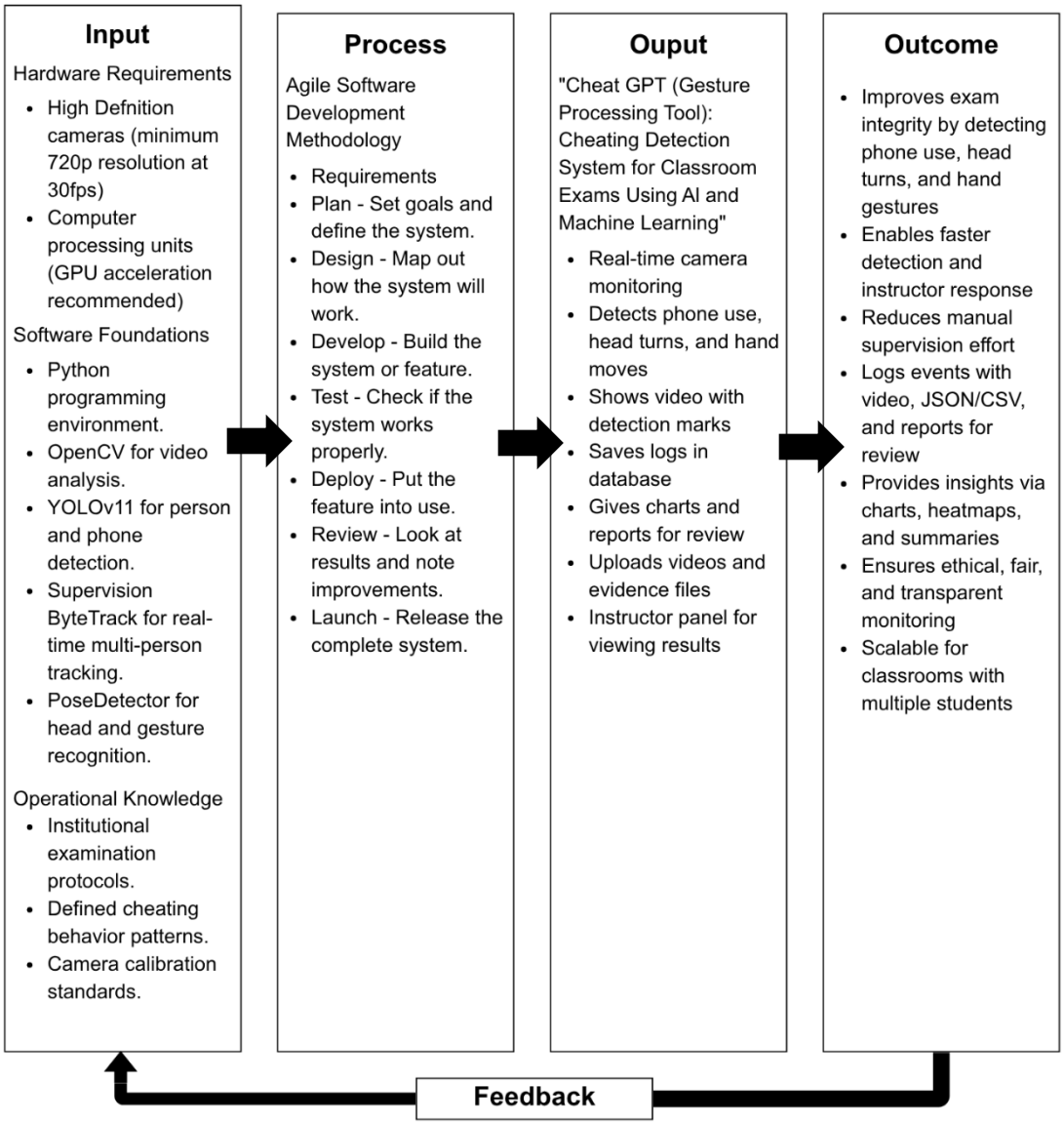
By combining computer vision tools with academic integrity models, the system aims to improve honesty during exams at CMU. It can detect head movements greater than 30 degrees and flag them for review, identify unauthorized items such as phones with up to 89% accuracy, and create a visible monitoring system that discourages cheating while promoting fairness. Integrating these capabilities, the system transforms ordinary classroom cameras into intelligent tools that not only monitor student activity but also promote equitable assessment practices, enhance instructor oversight, and reinforce a culture of academic honesty throughout the university.

**Conceptual Framework**

The system's architecture follows an Input-Process-Output-Outcome (IPOO) model to transform technological resources into cheating detection.

**Figure 1**

*IPOO model of the Cheat GPT system transforming video input into proctor alerts and detection logs.*




The conceptual framework in Figure 1 presents the Cheat GPT system using the Input-Process-Output-Outcome (IPOO) model, showing how it operates from start to finish. The input includes high-definition cameras (≥720p at 30 FPS),

GPU-enabled computers for real-time processing, and software tools such as Python, OpenCV, YOLOv11, PoseDetector, and Supervision ByteTrack for multi-person tracking and gesture recognition. Operational knowledge—including institutional exam protocols, defined cheating behaviors, and camera calibration standards ensures accurate interpretation of classroom activity.

The process follows an Agile Software Development Methodology, allowing incremental development and iterative improvement. Planning and design established system goals and workflow, while development focused on gesture detection, phone usage identification, and pose estimation. Testing with live and recorded sessions verified system performance, and each review cycle refined detection thresholds, overlays, and responsiveness. Deployment and launch provided a reliable, ethically aligned tool for classroom monitoring.

The output is the fully functional Cheat GPT (Gesture Processing Tool), enabling real-time detection of phone use, head turns, and hand gestures. The system provides video overlays, structured database logs, charts, and reports for instructor review, with functionality for uploading recorded sessions and evidence files. The instructor panel allows effective supervision and post-exam analysis.

The outcome demonstrates improved exam integrity, faster detection and instructor response, reduced manual supervision, and structured analytical insights via charts, heatmaps, and reports. Ethical monitoring ensures fairness and privacy, while system scalability allows deployment in classrooms with multiple

students. The video output serves as a reference, showing real-time overlays and event logs, confirming the system's reliability and practical applicability.

**Statement of the Problem**

Academic dishonesty during exams has always been a problem in schools, and City of Malabon University is no exception. Despite the presence of proctors to monitor students during exams and the strict implementation of university rules, certain students still manage to violate those rules without being caught. Whether it involves using smartphones, concealing cheat sheets, or utilizing discreet gestures, these behaviors frequently remain unnoticed. This happens particularly in classrooms where a single proctor oversees numerous students. It is very challenging for one individual to observe every student taking an exam simultaneously. As a result, there is an evident requirement for a system that can assist proctors in overseeing students more efficiently.

This research seeks to answer the following questions:

1. How do the core components and sub-modules of the Cheat GPT system function collectively to detect classroom cheating gestures?
2. How effective is the Cheat GPT in promoting academic integrity by detecting suspicious gestures and device use in real-time exam conditions?
3. How do selected computer vision and machine learning algorithms perform in recognizing gestures and detecting device use under classroom conditions?

4. How does the Cheat GPT perform when evaluated against the ISO/IEC 25010 quality model (covering performance efficiency, reliability, and usability)?

**Assumptions**

The following assumptions are drawn from the scope and methodology of the study. These represent the conditions presumed to be true for the system to function as intended.

**A1:** The classrooms where the Cheat GPT system operates have stable conditions, including adequate lighting, proper camera placement, and minimal obstructions, to ensure accurate gesture and object detection during examinations.

**A2:** The system's pose estimation and object detection algorithms are capable of identifying suspicious gestures and unauthorized device usage with reasonable accuracy.

**A3:** The system is expected to produce a noticeable number of false positives, wherein ordinary student behaviors may be incorrectly flagged as suspicious.

**A4:** Students and faculty members participating in the study understand the ethical guidelines and have given informed consent before data collection. All participants are expected to behave naturally during testing and provide honest feedback during evaluation.

**A5:** All detections and alerts generated by the Cheat GPT system will be verified by the proctor before any decision is made. The system is designed to assist

proctors by providing evidence-based alerts rather than autonomously determining instances of cheating.

**Significance of the study**

This study presents Cheat GPT, an AI-powered system designed to support proctors and educational institutions in maintaining academic integrity during classroom examinations. By applying computer vision and machine learning, the system detects suspicious gestures and unauthorized device use. The findings and outputs of this study offer benefits to multiple stakeholders, prioritized from the most directly affected to the least.

- **For City of Malabon University (CMU):** CMU stands to gain substantial benefits from the system once it is fully deployed. The system will strengthen the university's commitment to academic integrity by providing automated and consistent monitoring during examinations. Its scalable structure also allows CMU to expand its implementation across various departments, including programs that administer practical or hands-on assessments.
- **For the College of Computer Studies (CCS):** The project demonstrates the department's ability to leverage emerging technologies to address practical academic challenges. Through rigorous testing and development, CCS students gain hands-on experience in system refinement, surveys, and performance evaluation, ensuring the tool is robust before full university adoption.

- **For Faculty Members:** Instructors benefit from reduced monitoring burdens during exams. The system automatically detects, logs, and reports suspicious behaviors, providing short video clips or annotated screenshots for review. This enables more accurate, evidence-based judgment while minimizing subjective errors.
- **For Students:** Cheat GPT promotes fairness by discouraging dishonest behavior and ensuring consistent examination conditions. Students benefit from a level playing field and are not pressured to report peers, reducing potential social discomfort.
- **For the Proponents:** Student developers gain practical experience in advanced technologies such as pose estimation, object detection, behavioral analysis, and AI ethics. This equips them with skills for future roles in software development, AI engineering, or applied research.
- **For Future Development:** With continued improvements, Cheat GPT may be adapted for hybrid or online exam environments. The system's deployment and refinement at CCS provide valuable feedback and data, supporting future scalability and ethical application in broader educational contexts.

**Scope and Delimitation**

This study will focus on the development of a real-time monitoring system intended to assist proctors during classroom examinations. The system will utilize

computer vision and machine learning to observe and analyze student gestures throughout examination sessions. It will be designed to detect indicators of academic dishonesty, such as frequent or abrupt head turning. The system will also capture suspicious body movements, including reaching under the desk, frequently extending hands, passing objects, or hiding their hands. Additionally, it will detect mobile phones, specifically actions like holding, placing it on the lap or desk.

The system will be intended for implementation within the College of Computer Studies (CCS) at City of Malabon University to enhance monitoring during examinations. Its primary purpose will be to support human proctors by serving as an additional surveillance tool during test sessions. The system will operate using only one camera installed in the classroom, positioned to cover the entire examination area within its field of view.

The study will be limited to face-to-face, physical classroom settings and will not apply to online or remote examinations. Detection will be confined to the camera's field of view and may not capture gestures occurring outside of the frame or subtle acts that do not involve observable movements, such as silent cues or the use of hidden notes. The system will employ face and body tracking, but will not include facial recognition features for identity verification.

External conditions such as poor lighting, obstructed views, or improper camera placement may affect the system's detection accuracy. It will not independently determine instances of cheating; rather, it will provide alerts and

snapshots of potentially dishonest behavior for human proctor review. The system will record the entire exam session and save flagged cheating events, depending on the device's storage capacity. Although it will be operable on a standard computer, optimal real-time performance is expected when run on a device equipped with a graphics processing unit (GPU).

**Definition of Terms**

- *Cheat GPT.* is the AI-powered cheating detection system developed in this study. It integrates computer vision and machine learning to monitor student gestures during exams in real time using a single camera, detecting gestures, posture, and unauthorized device use.
- *Gesture Recognition.* refers to the process by which the system identifies student body movements, such as hand raising or head turning, using pose estimation algorithms. Specific patterns are classified as potentially suspicious behaviors.
- *Pose Estimation* is a computer vision technique used to detect and track key body points, including the head, shoulders, and arms. This enables real-time analysis of body posture and movement to infer behavioral patterns during an exam.
- *Mobile Phone Detection* is the module responsible for identifying mobile phones or electronic gadgets, using pre-trained deep learning models to detect unauthorized items within the camera frame.

- *Evidence Logging.* is the process of capturing screenshots or short video clips of detected events, storing them with timestamps and detection details for faculty review or documentation in case of student appeals.
- *Real-time Notification.* refers to the system's immediate alerting mechanism that informs the proctor when a suspicious gesture is detected. Notifications appear as dashboard updates, on-screen prompts, or log entries for instant awareness and action.
- *Machine Learning*. A model is a trained algorithm used to analyze visual data, detect objects or gestures, and learn behavior patterns. Models such as YOLOv11, OpenPose, or MediaPipe are utilized for these tasks.
- *Academic Integrity*. refers to the fairness and honesty maintained during examinations. The system supports academic integrity by reducing undetected cheating and promoting equal conditions for all students.
- *Frame Per Second*. measures how many visual frames the system can process per second. Higher FPS indicates more responsive and efficient real-time detection during live exams.
- *You Only Look Once (YOLO)* is a real-time object detection algorithm implemented in Cheat GPT to efficiently detect students, phones, and other objects in each video frame. YOLOv11 provides both speed and precision suitable for live classroom monitoring.
- *Supervision ByteTrack* is a tracking framework that assigns persistent IDs to students across frames, allowing continuous monitoring even when

students move, overlap, or partially exit the frame. This ensures accurate behavior analysis over time.

# Chapter 2

# METHODOLOGY

This chapter focused on the research methodology related to the study. It described the research design, participants, sampling procedure, research instrument, data gathering procedure, data analysis and statistical treatment, ethical considerations, informed consent, and anonymity and confidentiality. The methodology provided a systematic framework for conducting the research, ensuring the reliability and validity of the findings.

## Research Design

This study employs a quantitative developmental research design to guide the systematic development, evaluation, and refinement of the Cheat GPT, a real-time cheating detection system intended for use during classroom examinations. Developmental research is appropriate for technology-oriented projects where the goal is to construct innovative systems through iterative prototyping and empirical testing. Developmental research emphasizes design cycles that support continuous improvement through implementation, analysis, and revision based on observed outcomes and stakeholder feedback (Richey & Klein, 2007). This approach allows researchers to not only test the technical performance of the system but also to assess its practical applicability and user acceptance in the classroom.

The methodology is structured into three main phases. The first phase,

system development, applies the agile software development methodology, wherein system components are incrementally built, tested, and refined across multiple cycles. Core modules such as YOLOv11 for pose estimation and object detection are developed using Python and OpenCV, with training data based on annotated gestures such as head turning beyond 30 degrees and unauthorized device usage (Bancud, 2021; Zuo et al., 2024).

The quantitative component will involve structured data collection, such as measuring detection accuracy (precision, recall, F1-score), processing speed (FPS), and user satisfaction via Likert-scale surveys. These metrics will validate the system's effectiveness under ISO/IEC 25010 standards for software quality. By combining developmental flexibility with quantitative rigor, this design ensures the Cheat GPT system is both empirically validated and practically viable for classroom deployment.

The use of a developmental design combined with agile methodology allows for continuous enhancement of system accuracy and responsiveness under varied classroom conditions. This approach is critical in minimizing false alerts and adapting to real-world constraints (e.g., occlusion, lighting), as emphasized in studies on real-time attention monitoring in classrooms (Trabelsi et al., 2023).

***Software Architecture***

**Figure 2**

*System Architecture of Cheat GPT*

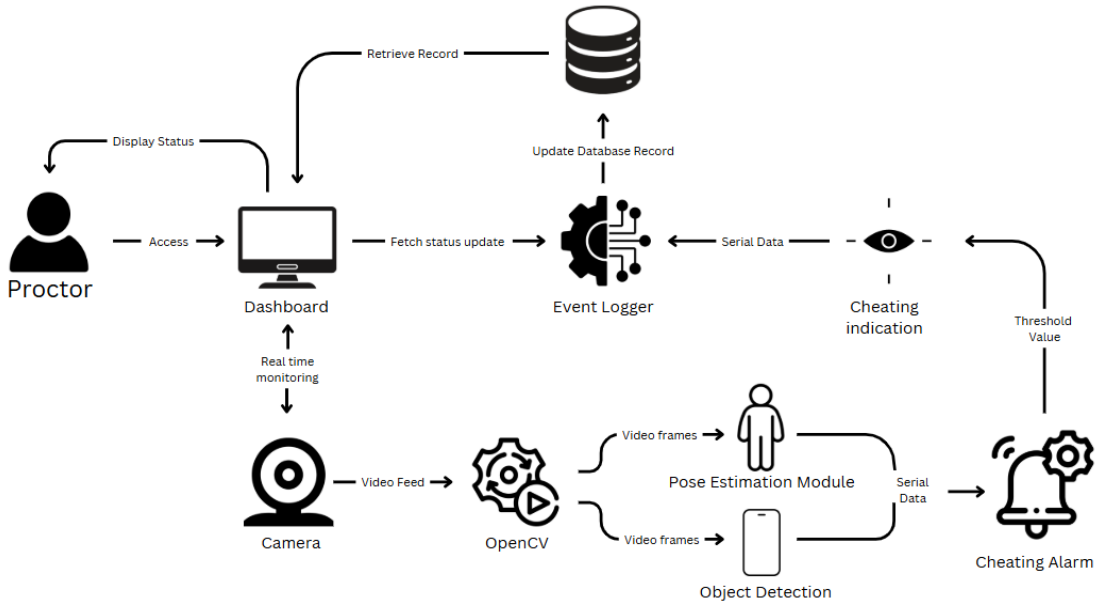



Figure 2 shows the system architecture of Cheat GPT, which monitors students during exams to detect possible cheating. It has five main components: data acquisition, preprocessing, detection, logging, and user interaction.

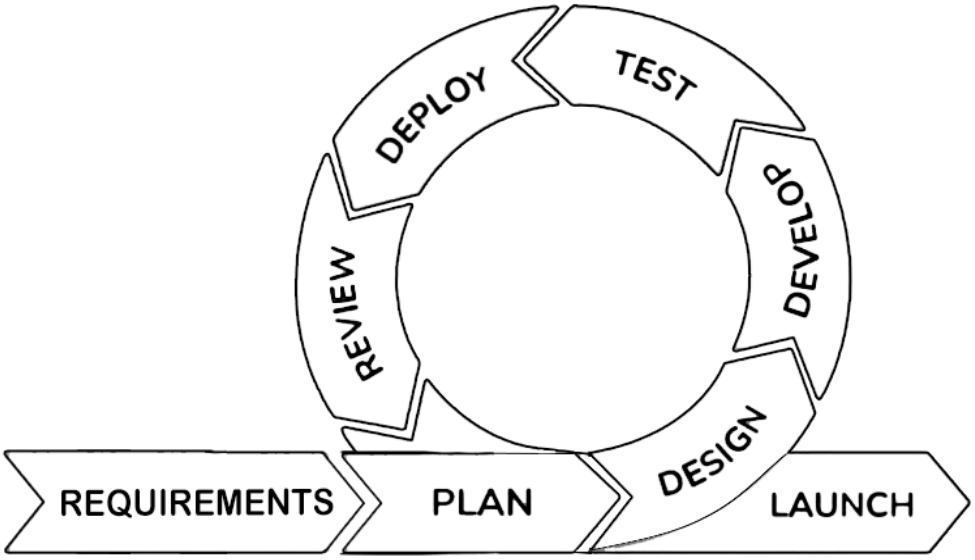
A webcam records live video, analyzed by two AI modules: Pose Estimation (posture/head movement) and Mobile phone detection. Detected data is sent to the Cheating Indication module, which flags suspicious behavior and triggers a notification if needed.

The Event Logger stores results and updates the Dashboard, allowing proctors to monitor, receive notifications, and review events. The system ensures efficient exam monitoring through AI and human oversight.

***Software Development Methodology***

**Figure 3**

*Agile Software Development Flow*




The Agile Software Development Methodology is used to guide the creation of the Cheat GPT. The Agile model supports building the system through small, manageable steps while allowing space for revision and improvement along the way. This approach was selected due to the evolving nature of the project's requirements and the need to test each component independently.

Unlike traditional methods that require fixed plans, Agile development provides flexibility to adjust and improve based on what works during testing. Since Cheat GPT is an offline, single-user application, it was important to design each module, such as gesture recognition, mobile device detection, and real-time notifications, in a way that could be tested and refined without waiting for the full system to be completed.

Using Agile also made it easier to update features, revise the entity-relationship model, and simplify system flow based on how the application was functioning in practice. For example, early plans for user logins and proctor details were removed once it became clear the system would run independently on the user's device.

Each development cycle focused on designing, implementing, and reviewing a specific feature. After each cycle, improvements were made before moving forward. This method helped reduce errors and allowed faster feedback on functionality. In the case of Cheat GPT, the Agile approach helped ensure that each part of the system was purposeful, efficient, and aligned with the overall goal of detecting cheating behavior using AI and video inputs.

***Data Flow Diagram***

The Data Flow Diagram (DFD) is structured into multiple levels to represent the system's functionality at varying degrees of detail. The Level 0 DFD, commonly referred to as the context diagram, provides a high-level abstraction of the entire system by illustrating it as a single process and outlining its primary interactions with external entities such as the proctor, camera, and record management system. The Level 1 DFD breaks down this overarching process into several sub-processes, offering a more detailed representation of the system's internal operations.

**Figure 4**

*Context Diagram of Cheat GPT*

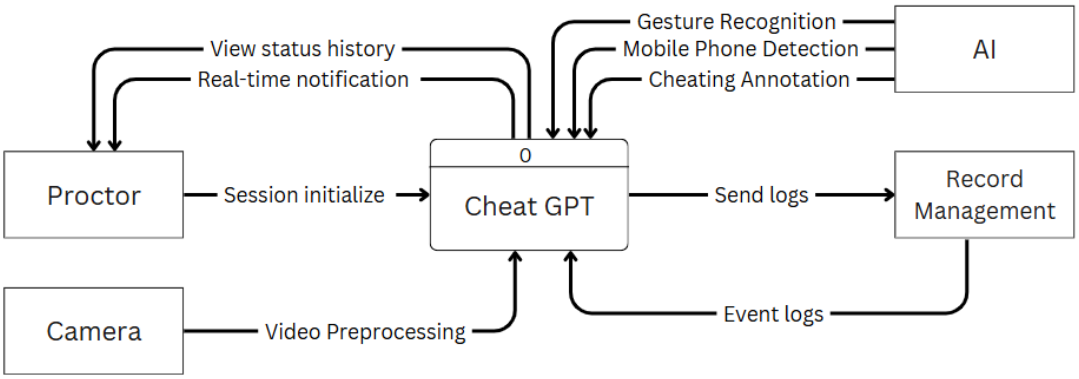
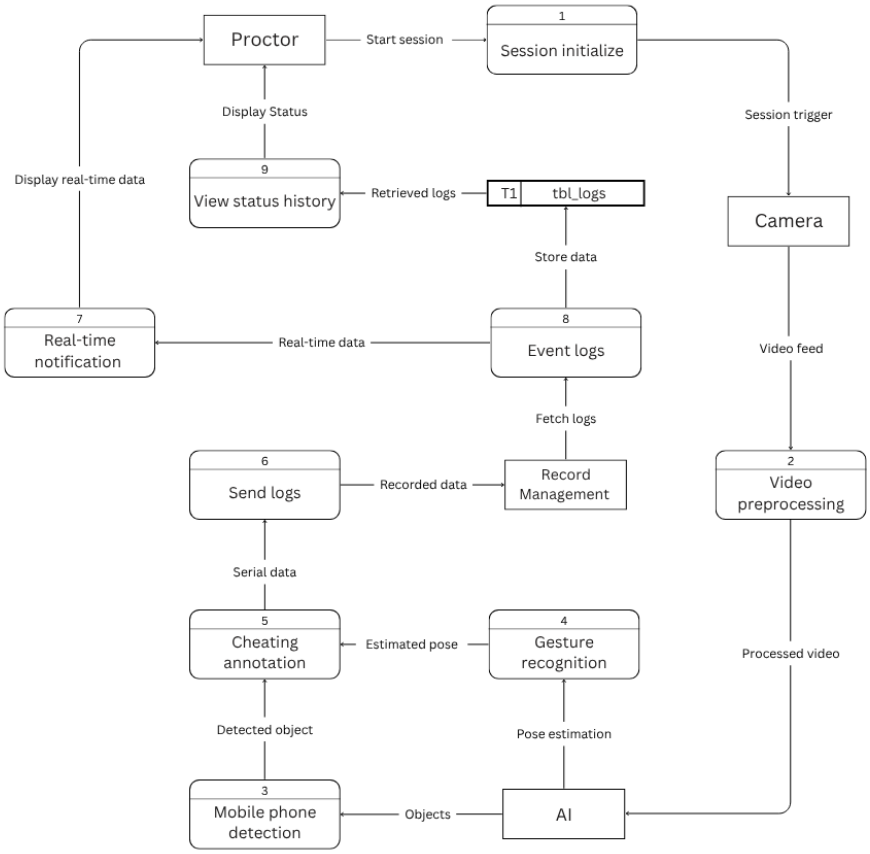



Figure 4 presents a high-level overview of the interactions between the Cheat GPT system and its external entities. This context diagram offers a simplified yet comprehensive depiction of how the system operates within a classroom examination environment. It begins by receiving live video input from a camera feed and examination-related session data from the proctor. These inputs are then processed by AI-powered modules responsible for gesture recognition and mobile phone detection. Based on the analysis, the system interprets student gestures, flags suspicious actions, and sends real-time alerts to the proctor. Simultaneously, relevant event data is logged and managed through the record management system, enabling post-exam retrieval and review. Key data flows include session initialization, video preprocessing, detection interpretation, alert generation, and log access. Altogether, the diagram highlights the essential components and data exchanges required to support accurate and responsive cheating detection during classroom examinations.

**Figure 5**

*Data Flow Diagram Level 1 of Cheat GPT*




This figure illustrates the Level 1 Data Flow Diagram of the Cheat GPT system, providing a detailed view of how data moves through the main components of the cheating detection process during classroom examinations. The process begins when the proctor logs into the system and initiates a session, which in turn activates the camera to capture live video of the examination environment. The captured video frames are then streamed into the system and processed in real time by multiple AI modules. These modules include object

detection algorithms, such as YOLOv11, for identifying students and unauthorized devices, and a pose estimation model for analyzing gestures, head movements, and posture.

In addition to real-time notifications, all detected events are systematically recorded and stored in a structured database, including relevant metadata such as timestamps, session information, and video clips of the incident. This logging functionality ensures that faculty members can retrieve records, review evidence, and generate reports for post-exam analysis or student appeals.

The Level 1 DFD thus highlights the flow of data from input, through processing, to output, emphasizing the integration of AI-driven analysis, automated logging, real-time alerting, and instructor review. This comprehensive flow ensures accuracy, accountability, and transparency in monitoring classroom exams while supporting academic integrity.

Furthermore, it demonstrates how the system seamlessly integrates advanced AI technologies with instructional oversight, enabling proctors to monitor multiple students effectively, ensure fairness, maintain academic integrity, and provide reliable evidence for review and decision-making in classroom examinations.

***System Flowchart***

**Figure 6**

*System Flowchart of Cheat GPT*

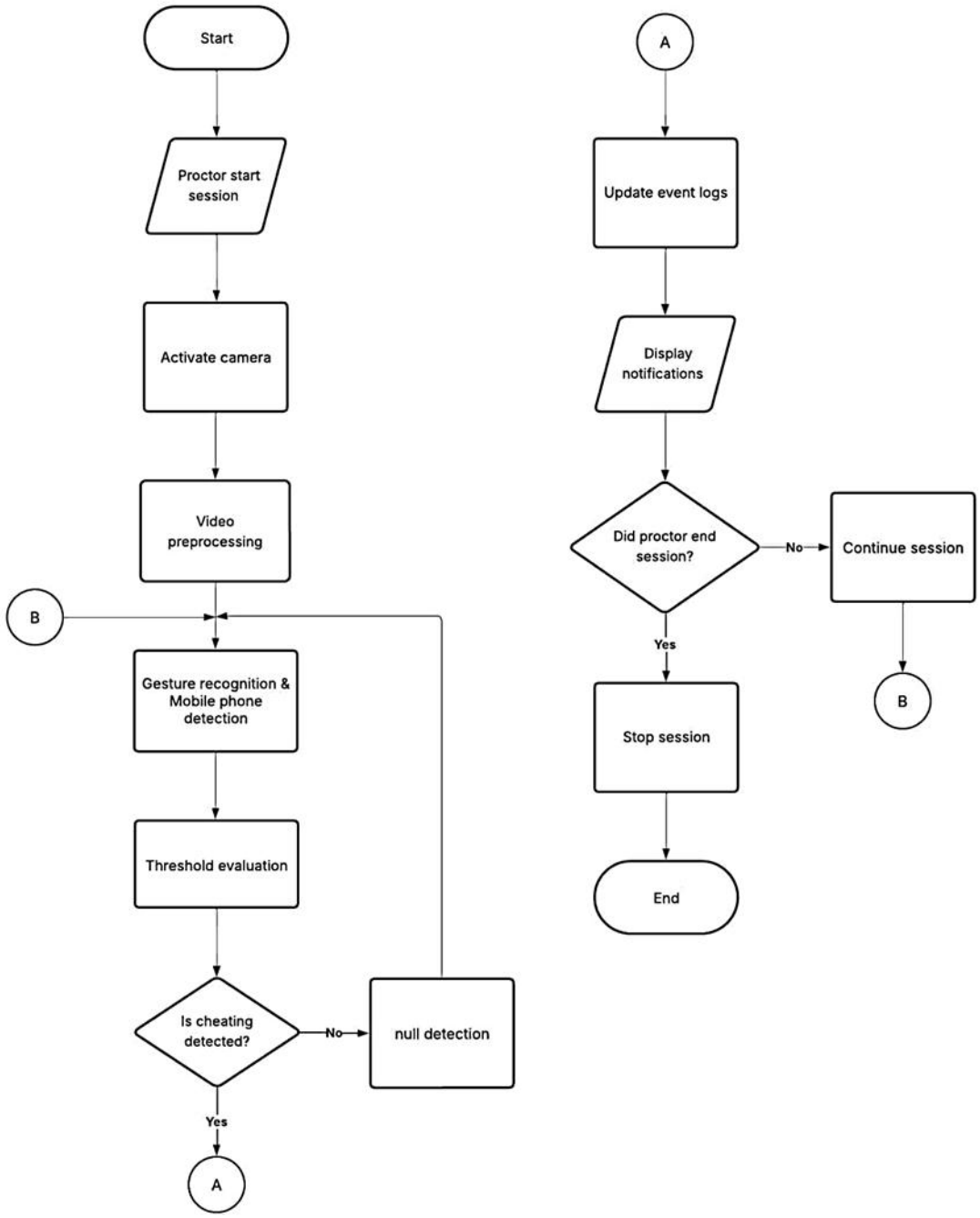


Figure 6 presents the complete operational workflow of Cheat GPT. Upon activation, the system triggers the webcam to continuously capture live video of students during the examination.

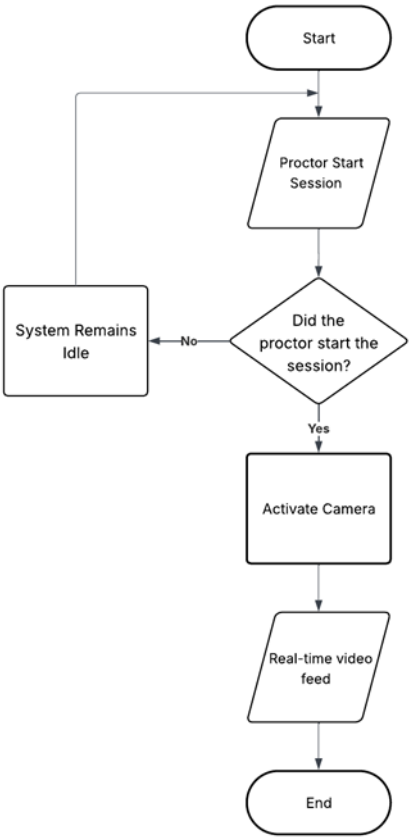
Each captured frame is passed into the video preprocessing, where it is resized, normalized, and converted into a suitable format for AI analysis. The cleaned frame is then fed into two core AI modules: the Gesture Recognition and Object Detection module. The results from these modules are processed by the Cheating Evaluation Engine, which checks them against predefined thresholds and behavior rules. If the system determines that a cheating incident is likely, it triggers three simultaneous actions: (1) it logs the event to the detection database along with supporting evidence (timestamp, detection type, and media file); (2) it sends a real-time notification to the proctor's dashboard to enable immediate human oversight; and (3) it updates the session's status history for auditing purposes.

The monitoring process continues in a loop, capturing, analyzing, evaluating, and responding until the proctor ends the session. Upon termination, the system halts all processing modules and compiles a complete session log, which may include exported reports of all flagged incidents and annotated feedback. This end-to-end system structure supports continuous, AI-assisted exam surveillance while maintaining data integrity, real-time responsiveness, and transparency for academic review.

**Program Flowchart**

**Figure 7**

*Session Initiation of Cheat GPT*




This flowchart illustrates the workflow for initializing a proctoring session in the Cheat GPT. The process begins when the proctor selects the “Start Session” option, which activates the webcam to begin the live video feed. The system verifies that the camera is functioning and that the video stream is active, ensuring readiness for real-time monitoring. Once the feed is confirmed, the session continues until manually concluded by the proctor, marking the formal start of exam surveillance.

**Figure 8**

*Video Preprocessing of Cheat GPT*

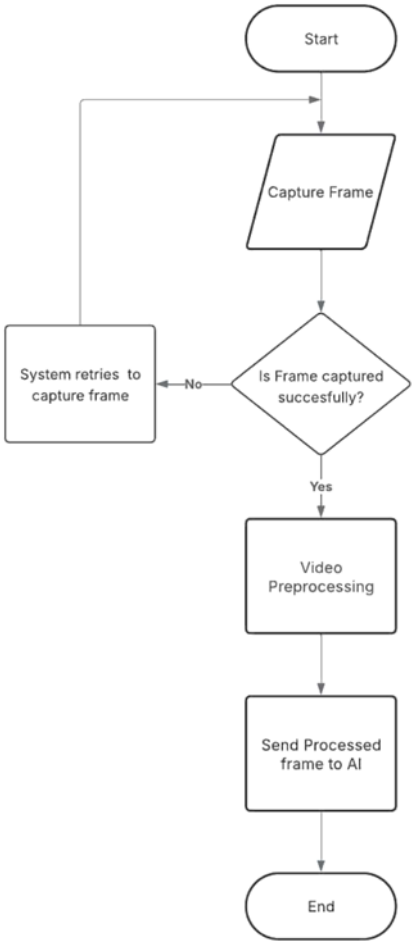
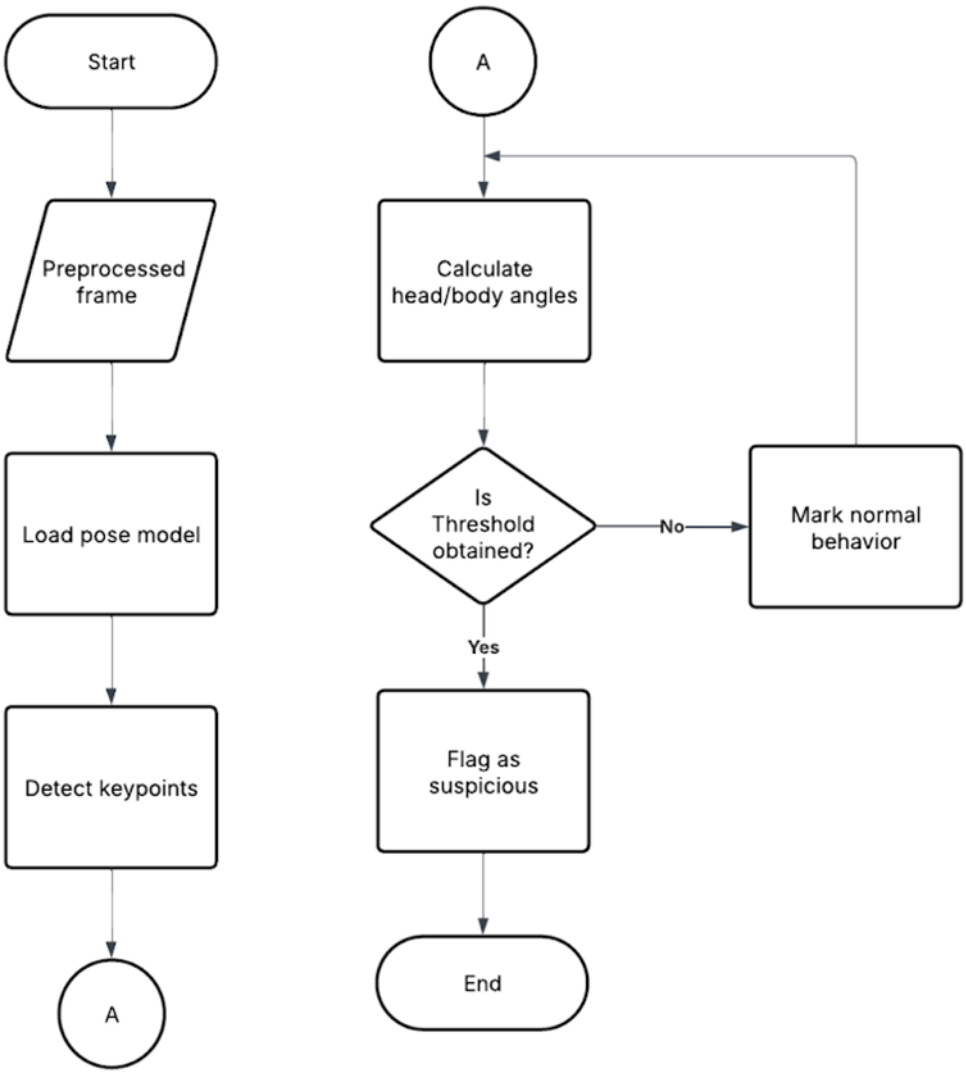



Figure 8 outlines the workflow of video frame handling within the Cheat GPT. The process begins with the system capturing a live frame from the camera. The frame is then passed through the video preprocessing module, where operations such as resizing, normalization, and color correction are performed for analysis. Once preprocessing is complete, the cleaned frame is forwarded to the

AI components: Pose Estimation and Object Detection, which handle real-time cheating detection.

**Figure 9**

*Gesture Recognition of Cheat GPT*



This flowchart represents the internal logic of the gesture recognition within Cheat GPT. The process begins with loading the pose estimation model, such as YOLOv11. From there, the system detects key body landmarks, particularly the head and shoulders, and calculates angles to assess student posture and movement. These angles are then compared against predefined thresholds. If the angles fall below the threshold, the system continues monitoring. If the threshold is exceeded, the gesture is flagged as suspicious and marked for further evaluation.

**Figure 10**

*Mobile Phone Detection of Cheat GPT*

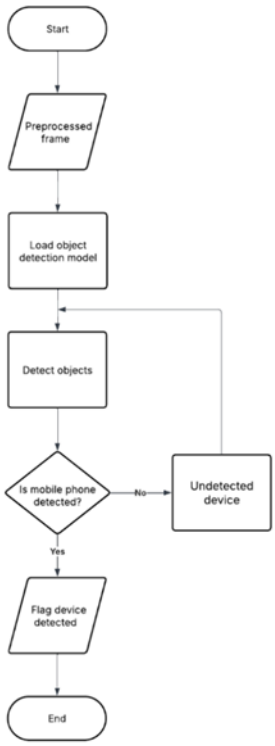
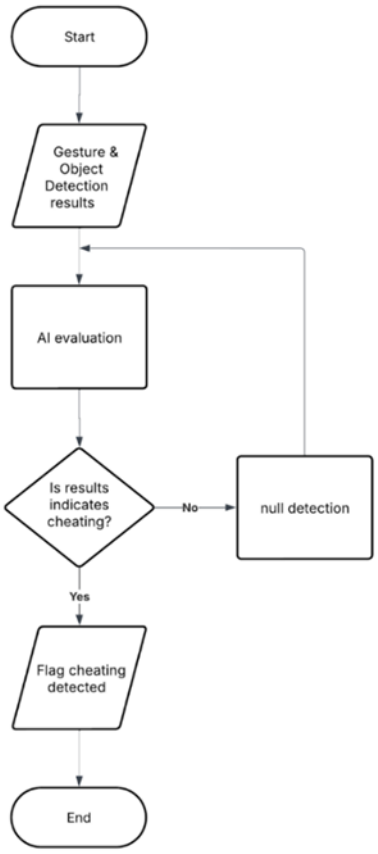


Figure 10 outlines the workflow of the mobile phone detection module in the Cheat GPT system. The process begins by loading the pre-trained object detection model, such as YOLOv11. Once the frame is ready, the system initiates the detection process of physical objects within the frame. If a mobile phone is detected with a high confidence score, it is flagged as a violation and recorded for further evaluation. If no phone is found, the system classifies the frame as clean.

**Figure 11**

*Cheating Annotation of Cheat GPT*



This flowchart presents the decision-making logic used by the Cheat GPT to annotate cheating gestures based on AI analysis. The process begins with receiving outputs from the gesture recognition and object detection modules. These results are passed to the AI evaluation engine, which assesses the data against a predefined set of cheating conditions, such as excessive head turning or mobile phone presence. If the criteria for cheating are met, the system flags the incident and records it accordingly. If the results do not meet the cheating threshold, the detection is marked as null.

**Figure 12**

*Real-time Notification of Cheat GPT*

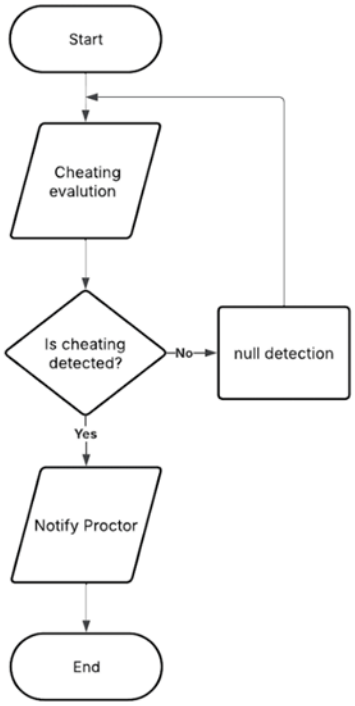
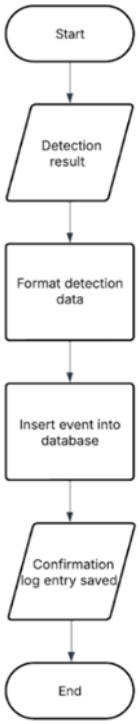


Figure 12 details the real-time cheating notification process in the Cheat GPT. The process begins with the system evaluating gesture and object detection results to determine whether a cheating gesture is present. If cheating is detected, the system immediately triggers a notification to the proctor via the dashboard, enabling timely human oversight during the exam. If no suspicious gesture is found, the system confirms a null detection. Regardless of the result, the system performs a cheating evaluation to log and review the detection outcome, then loops back to continue monitoring. This flow ensures continuous AI-driven surveillance with instant alerts to support academic integrity in real time.

**Figure 13**

*Event Logs of Cheat GPT*



This flowchart outlines the event logging process. The process begins immediately after the system detects a suspicious gesture or unauthorized object. The raw detection data, such as event type, timestamp, and confidence score, is formatted into a structured record. This record is then inserted into the detection log database, ensuring it is properly stored for future review. Once the event is logged, the system saves a confirmation entry to verify that the operation was successful. The process ends after the event has been securely recorded, maintaining the system's accountability and traceability.

**Figure 14**

*Send Logs of Cheat GPT*

This flowchart illustrates the process by which the proctor retrieves detection logs from the Cheat GPT system. The sequence begins when a log access request is initiated, typically from the dashboard. The system then validates any provided filters, such as session ID, date range, or detection type, to ensure they are correctly formatted and complete. Once validated, the system constructs and executes a database query using the specified filters. Matching log entries are then retrieved from the database and made available for review. The process concludes after the logs are successfully fetched, supporting post-exam auditing and system transparency.

**Figure 15**

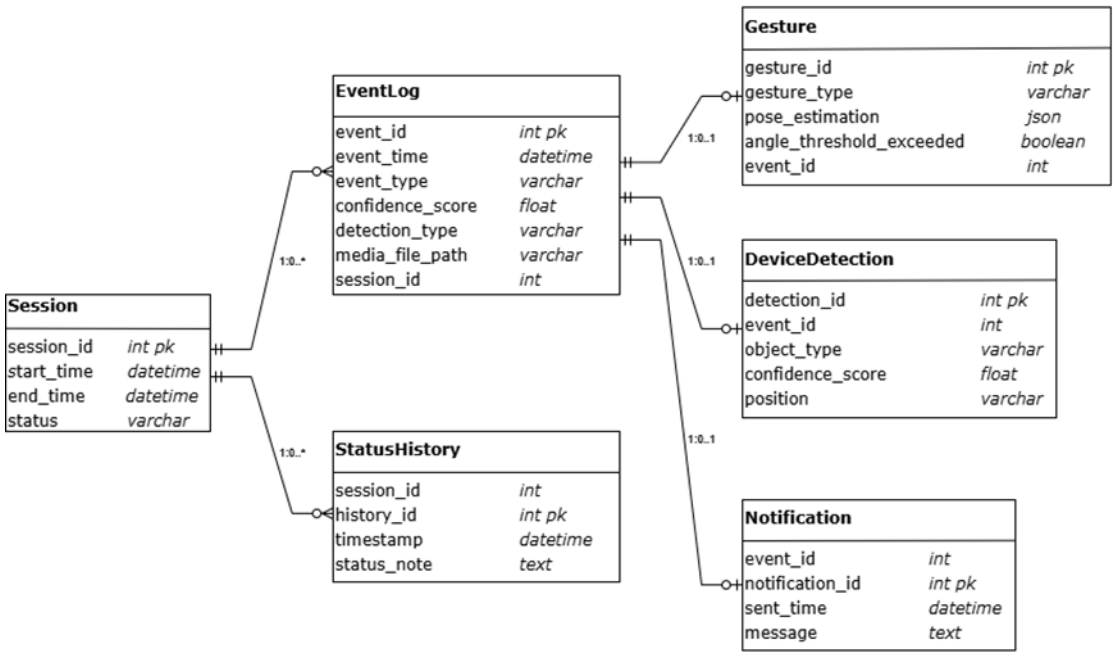
*View the Status History of Cheat GPT*

Figure 15 presents the steps involved in retrieving a complete session history within the Cheat GPT system. The process begins when the proctor selects a specific exam session for review. The system then accesses the session's status history records and retrieves all associated alerts, feedback from previous detections, and corresponding timestamps. Once compiled, this information is presented in an organized format via the dashboard, allowing the proctor to review the session's full behavioral log.

**Entity Relationship Diagram**

**Figure 16**

*Cheat GPT Entity Relationship Diagram*




The Entity Relationship Diagram in Figure 16 outlines the core structure of the Cheat GPT system's database, showing how key components interact with one

another during proctoring sessions. It uses a relational model to connect real-time monitoring events with gesture recognition, device detection, status updates, and system-generated notifications.

At the far left of the diagram is the Session table. A session refers to an individual exam monitoring instance. Each session has a unique session_id, along with start_time, end_time, and an overall status. This table acts as the root of the system's logging structure.

The relationship between Session and EventLog is 1:0..*. This means a single session may generate zero or many events depending on activity. Each event belongs to exactly one session (1), making this link essential for contextualizing all captured data.

Similarly, the 1:0..* relationship from Session to StatusHistory tracks changes in session state over time. A session may have multiple status updates (e.g., idle, active, flagged), each with a timestamp and status_note.

The EventLog table acts as a parent to three specialized detection tables. Each row in this table represents a captured event — such as a gesture, a detected object, or a condition that may trigger a system notification. Every event is linked back to a session via the foreign key session_id.

The relationship here is 1:0..1, meaning an event may contain gesture information, but not always. This depends on whether the event type is categorized as a gesture. If it is, the corresponding gesture_id and detailed pose data are stored in the Gesture table.

Like gestures, device-related events also use a 1:0..1 relationship. If an object such as a mobile phone is detected, it is logged in the DeviceDetection table along with its type, position, and confidence level. If no device is present, the link remains empty.

Notifications are messages triggered by specific events, such as repeated suspicious behavior. This relationship is also 1:0..1, indicating that not every event leads to a notification, but each notification must be tied to one specific event.

In this system, the “1” cardinality indicates required participation. For example, each event must be linked to one session. However, gesture, device, and notification entities are all optional extensions of the base event, determined by the type and context of each occurrence.

This ERD structure provides a modular and extensible design suitable for real-time applications. It ensures data integrity while maintaining the flexibility needed for different detection outcomes. Each relationship reflects a real-world monitoring logic, allowing the Cheat GPT system to accurately log, interpret, and respond to suspicious activity in an organized and scalable way.

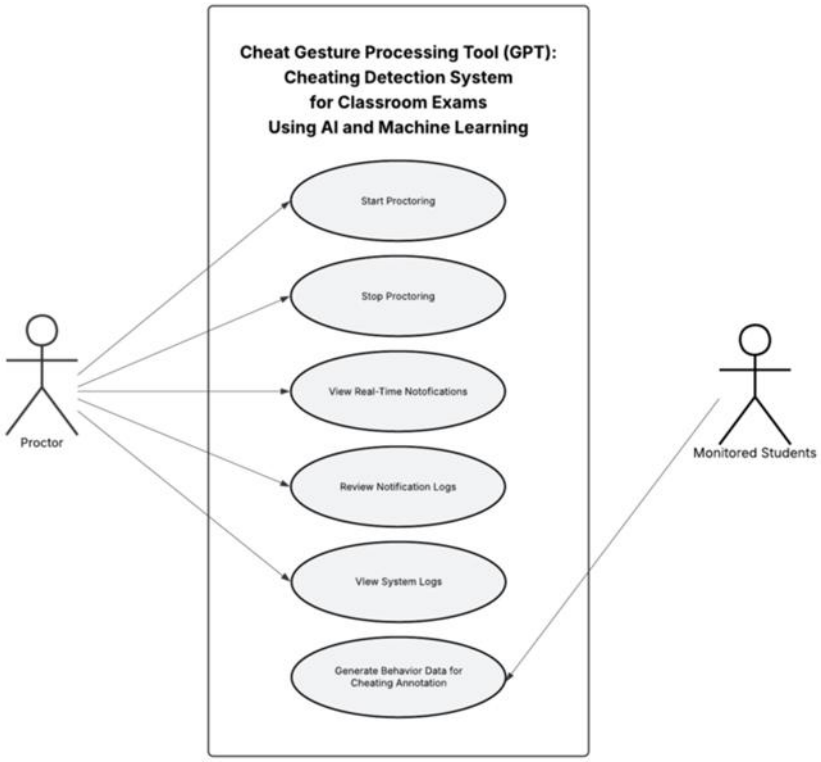
In conclusion, the Entity Relationship Diagram establishes a clear, modular, and relational structure that supports the real-time operations of the Cheat GPT system. By organizing data through well-defined relationships between sessions, events, gestures, devices, and notifications, the ERD ensures data integrity, consistency, and scalability. This design not only enables efficient tracking of exam

activities but also allows the system to be easily extended for future enhancements such as multi-camera integration or advanced analytics.

***Use Case Diagram***

**Figure 17**

*Cheat GPT Use Case Diagram*




This use case diagram illustrates the core functionalities of the Cheat Gesture Processing Tool (GPT). The diagram presents the interactions between the system and the proctor, who represents the faculty members from the College of Computer Studies. As the primary actor and system administrator, the proctor is responsible for managing and overseeing the system's operations, such as starting and stopping proctoring sessions, viewing real-time notifications,

reviewing alert logs, and accessing system logs. This key function ensures that the proctor maintains full control over the examination session, transparency and monitoring of the system, and monitors student behavior effectively using AI and machine learning technologies.

In addition to the proctor, the student is also included as a secondary actor. While students do not interact with the system directly, their physical behavior during classroom examinations serves as essential input for the AI. Through their movements and actions, students generate data that the system observes, processes, and compares against predefined thresholds to determine if a certain gesture may be considered suspicious or indicative of academic dishonesty. This indirect interaction plays a crucial role in enabling the system's gesture-based cheating detection capabilities.

**Population and Sampling Procedures**

In this study, the purposive sampling technique was used to select respondents based on specific criteria relevant to the objective of the research. The population of interest was the College of Computer Studies with fifteen (15) members, comprising seven (7) full-time and eight (8) part-time professors. Respondents were selected on the basis of their experience and involvement with and understanding of technology and AI innovation in the Information Technology field. The non-probability sampling technique employed enabled researchers to obtain information from those who were most likely to give well-informed answers

and valuable insights, ensuring that the data obtained were informationally relevant.

**Research Instruments**

The development of Cheat GPT, an AI-powered cheating detection system for classroom exams, requires an effective way to assess the performance efficiency and reliability of the system. This study employs a structured survey questionnaire to collect subjective user feedback data, aligning with the quantitative developmental research design of this study. The evaluation was also guided by the ISO/IEC 25010 software quality standard, with a particular focus on the attributes of usability, performance efficiency, and reliability.

The objective of the survey was to gather user feedback regarding their interaction with the system. The questionnaire, consisting of 16 questions, was organized into five main parts, namely usability, performance efficiency, reliability, usefulness, and ethical considerations. Each item was rated using a four-point Likert scale ranging from 1 - strongly disagree to 4 - strongly agree.

By integrating subjective user assessments with empirical performance data, the researchers were able to conduct a comprehensive evaluation of Cheat GPT. This approach enabled them to determine the system's potential to support academic integrity in classroom examination settings and to identify enhancements necessary for its broader implementation.

**Data Gathering Procedures**

With the consent and guidance of the research adviser, the researcher interviewed selected respondents consisting of seven (7) full-time and eight (8) part-time professors from the College of Computer Studies, for a total of fifteen (15) respondents. The use of the Cheat Gesture Processing Tool (GPT): Cheating Detection System for Classroom Exams Using AI and Machine Learning was viewed through purposive sampling. The survey used the Likert scale where 1 is 'Strongly Disagree,' 2 is 'Disagree,' 3 is 'Agree,' and 4 is 'Strongly Agree.'

The survey aimed to assess whether the system is useful and usable, how simple it is to use and understand, how dependable it is to use without interruption or error in advance and during the process of examination proctoring, as well as how precise it is, especially if the cheating detection system works well and accurately detects cheating behavior during the test. The information gathered from the participants will be applied to determine the effectiveness of the system and to guide updates for future use. pertinent to comment, and relevant to assessment, thereby enabling comprehensive and accurate results.

The respondents will be addressed through a survey tool in the form of closed-ended questions prepared on a four-point Likert scale ranging from "Strongly Disagree" (1) to "Strongly Agree" (4). The questionnaires will be completed with the help of the researchers, who will read the instructions aloud both before and during the response being filled. In the spirit of maintaining the confidentiality and security of the responses, the respondents' answers will be kept anonymous and undisclosed.

**Ethical Considerations**

This research seeks to improve fairness, safeguard student rights, uphold academic integrity, and ensure transparency alongside data security. Before data collection, crucial permissions will be secured from local authorities, and all required information will be compiled.

The students will be fully informed about how the system works, what data is being collected, and their rights. The system is designed to detect suspicious behavior, but the final decision will always be made by a proctor reviewer to prevent wrongful accusations.

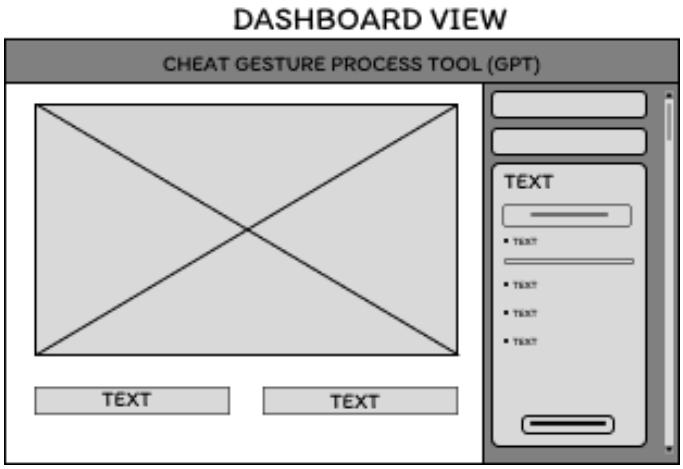
This research involves both software and hardware, focusing solely on gathering data pertinent to cheating detection, including smartphones, cheat sheets, facial expressions, and hand movements. All information will be encrypted and securely stored, with access restricted to authorized individuals only. After the academic year, all data will be permanently deleted.

***Prototype Testing***

To evaluate the usability and functionality of the Cheat Gesture Processing Tool (GPT), wireframes were created for additional modules and tested with intended users to gather feedback. The prototype demonstrated key interface components such as the dashboard, history, and recorded footage.

**Figure 18**

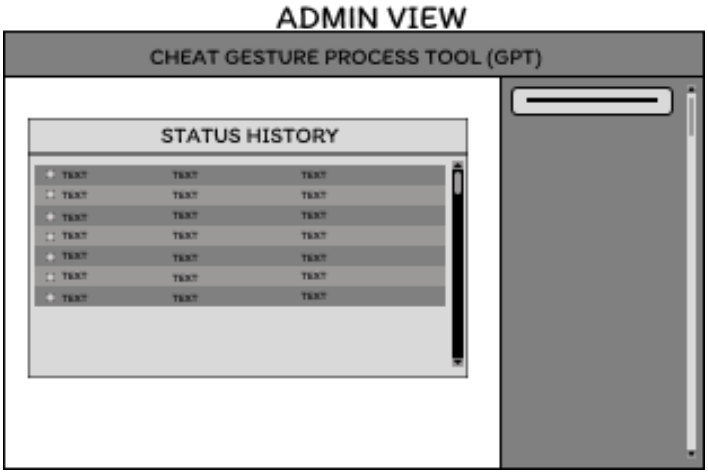
*Dashboard (Home) of Cheat GPT*




This page serves as the main interface for proctors. It contains the various modules, including Status history. It provides quick navigation and real-time updates, helping proctors efficiently monitor and review the examination sessions.

**Figure 19**

*Dashboard (History) of Cheat GPT*

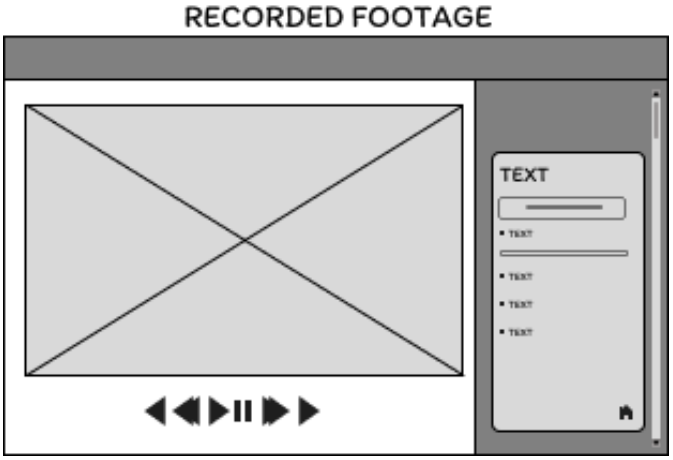



This interface is designed specifically for the proctor. Proctor can review, re-watch, and save the videos. It includes a labeled table displaying date, timestamps,

and event logs. The wireframe was tested to ensure that functions like re-watch and save work.

**Figure 20**

*Recorded Footage of Cheat GPT*




This is the page where the proctor can examine and re-watch the recorded videos. It features a large video player, navigation controls (play, pause, forward, rewind).

**Data Analysis and Statistical Tool**

To ensure the validity of the data, the information collected was organized and categorized methodically. Descriptive analysis was used to elaborate on the findings obtained from the questionnaires filled out by the participants for the researchers. Statistical method used to obtain the findings.

The weighted mean was used to analyze the data collected by the researchers on reliability, usability, accuracy, and effectiveness of the Cheat

Gesture Processing Tool (GPT): Cheating Detection System for Classroom Exams Using AI and Machine Learning.

By using the formula where x represents the weighted mean, w refers to the frequency or the number of respondents who selected a particular option, and x is the weight value, which corresponds to the numerical scale assigned to each Likert option (4 for *Strongly Agree*, 3 for *Agree*, 2 for *Disagree*, and 1 for *Strongly Disagree*). The term ∑ (w × x) refers to the total weighted score, which is obtained by multiplying the frequency of each response by its corresponding weight and then adding all of these products together. The denominator, w, is the total frequency, which simply refers to the total number of respondents.

**Table of Analysis and Interpretation of Data**

The Likert Scale Interpretation Table serves as a guide for analyzing and understanding the results of the data collected from the respondents.

**Table 1**

*Likert Scale Interpretation Table*

| Scale | Scale Range | Meaning |
|---|---|---|
| 4 | 3.26 - 4.00 | Strongly Agree |
| 3 | 2.51 - 3.25 | Agree |
| 2 | 1.76 - 2.50 | Disagree |
| 1 | 1.00 - 1.75 | Strongly Disagree |

The Likert Scale Interpretation Table is used to better understand the answers of the respondents. In this 4-point scale, a score of 3.26 to 4.00 means the respondents Strongly Agree with the statement, while a score of 2.51 to 3.25 shows they Agree. A score between 1.76 and 2.50 indicates they Disagree, and the lowest range of 1.00 to 1.75 means they Strongly Disagree. This table makes it easier to see the overall opinions or perceptions of the respondents.

# Chapter 3

## RESULTS, ANALYSIS, AND DISCUSSION

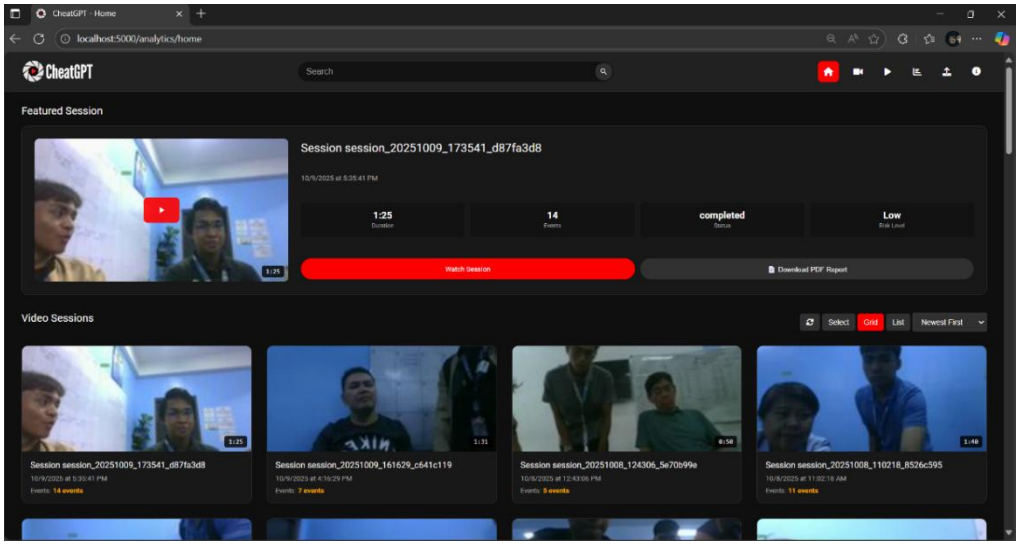
This chapter presents the results and discussion of the study's findings. It provides a detailed account of the data gathered and the statistical analysis performed. The results are presented in a clear and organized manner, followed by a discussion of their implications, relating the findings to the research questions and the theoretical framework. The chapter also compares the findings to existing literature, highlighting similarities, differences, and unique contributions to the body of knowledge.

### Interface Screenshots

This section presents the visual layout of CheatGPT, highlighting the user interfaces across various system pages. Each screenshot is accompanied by a brief description to showcase the functionality of monitoring, analytics, reporting, and management modules. The visuals emphasize the system’s professional design, real-time detection capability, and intuitive controls that ensure efficient classroom monitoring and analysis. These interfaces also demonstrate how instructors can easily navigate between live monitoring, historical data, and detailed reports for effective decision-making.

**Figure 21**

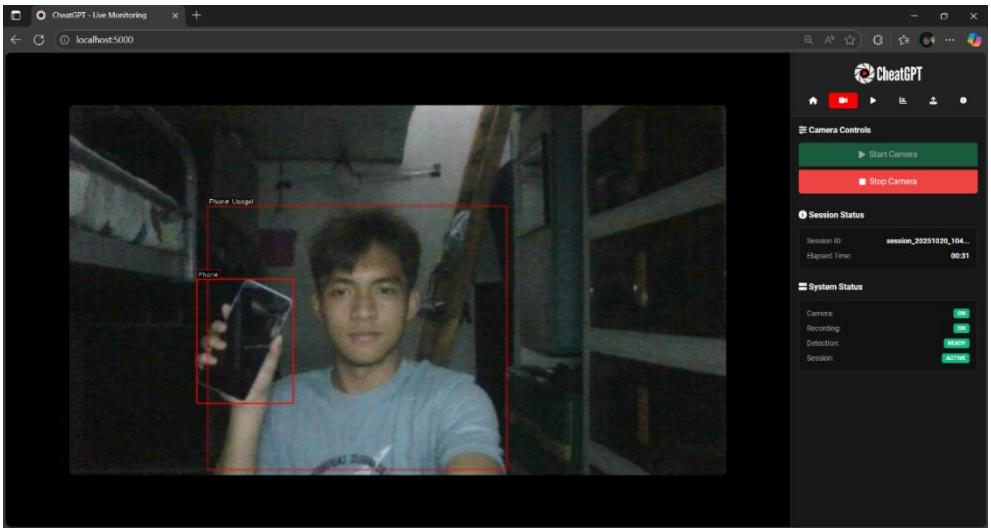
*Home Page of Cheat GPT*




The Home Page serves as the central dashboard for all classroom monitoring sessions. It displays recent videos, quick statistics, and tools for sorting, viewing, and managing recorded analytics efficiently.

**Figure 22**

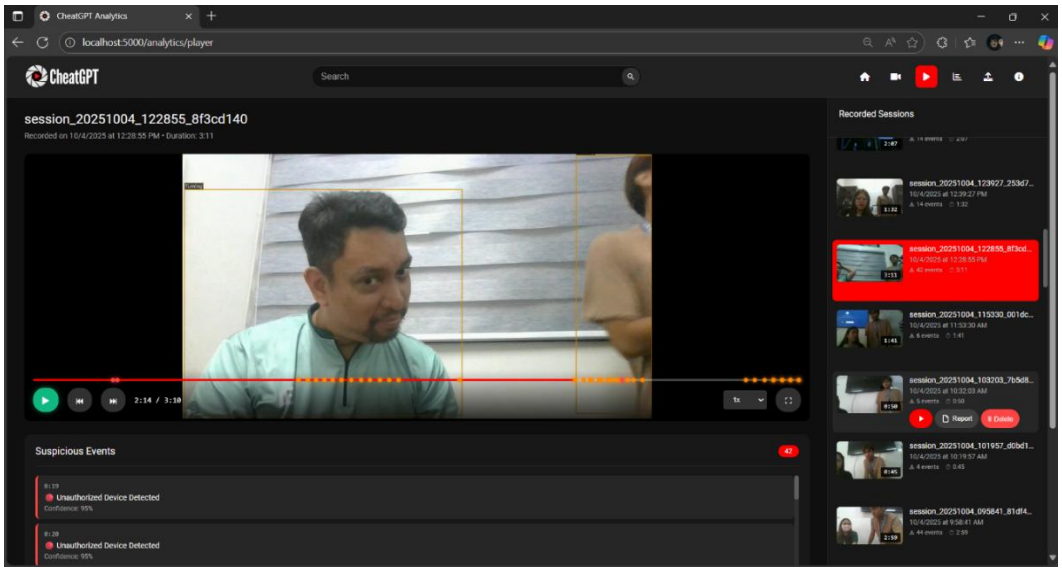
*Live Monitoring Page of Cheat GPT*



The Live Monitoring Page provides real-time detection and behavior tracking through the system's camera. Users can start or stop recording and monitor alerts instantly during live sessions.

**Figure 23**

*Player Page of Cheat GPT*




The Player Page features a custom video player with interactive event markers and playback tools. Users can analyze behavior patterns, navigate through events, and view overlays for detected activities. The player also allows users to pause, rewind, and go full screen on specific incidents for closer inspection. Additionally, summary statistics and event timelines are displayed alongside the video to provide context and support quick decision-making.

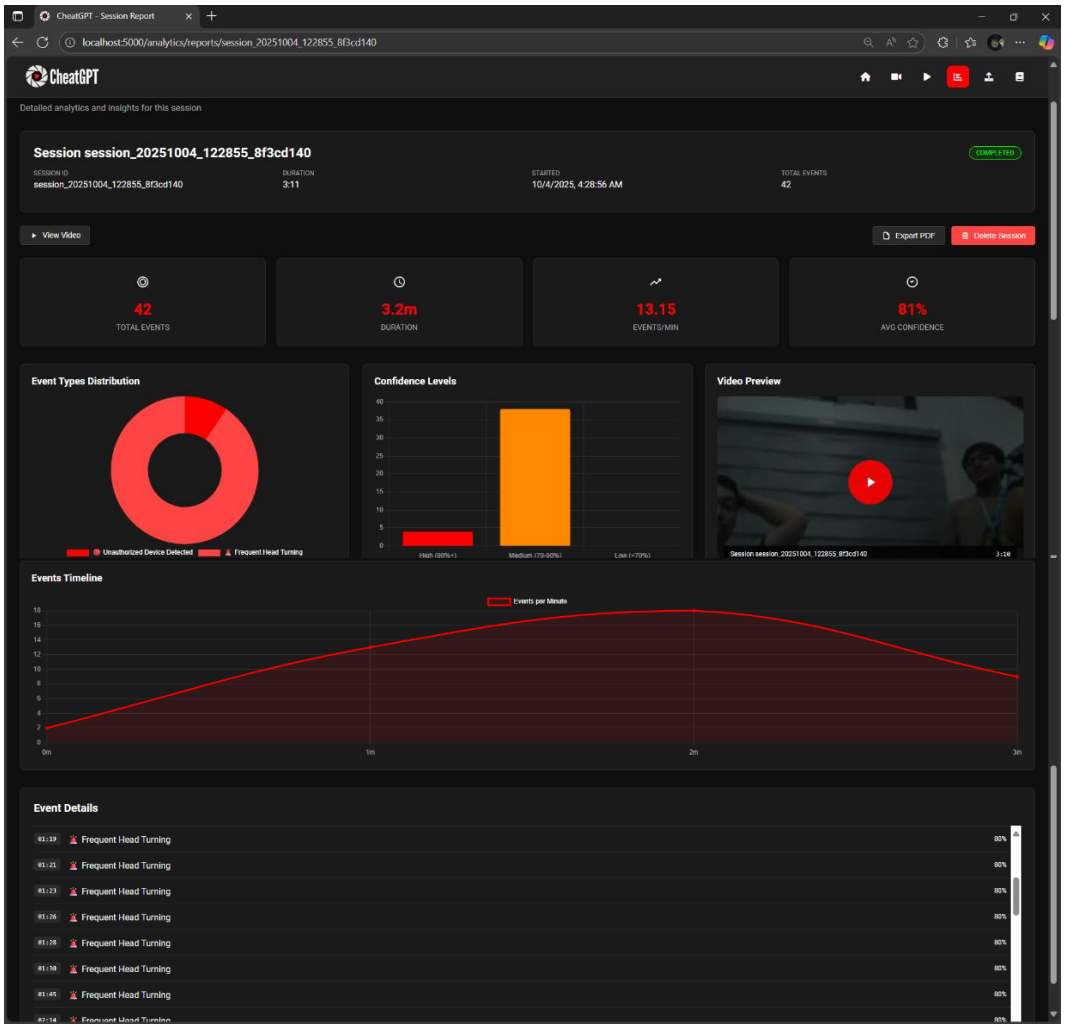
**Figure 24**

*Reports Page of Cheat GPT*

The report page lists all recorded sessions with advanced filters and batch processing options. It enables users to generate, compare, and export reports for deeper insights into classroom activity.

**Figure 26**

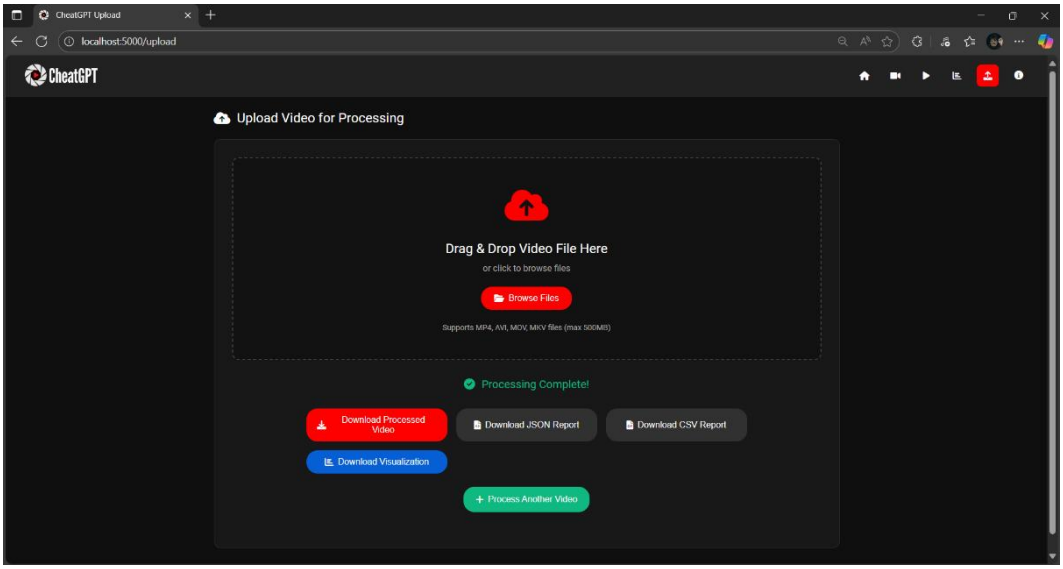
*Session Report Page of Cheat GPT*




The Session Report Page presents detailed analytics for a specific monitoring session. It includes a video preview, event timeline, charts, and exportable reports for documentation or review purposes.

**Figure 28**

*Upload the Page of Cheat GPT*




The upload page allows users to upload and process multiple classroom videos simultaneously. It supports drag-and-drop uploads, progress tracking, and automatic processing for analysis integration.

***Description of Modules***

This section presents the core modules and submodules of the CheatGPT System, an AI-powered classroom monitoring platform designed to detect, analyze, and record potential cheating behaviors in real-time. Each module fulfills a specific role that contributes to the system's overall efficiency, responsiveness, and reliability.

***Detection Engine Module***

This module serves as the system's core intelligence, responsible for real-time behavioral analysis through computer vision and rule-based interpretation. It

processes continuous video streams using object and pose detection models to identify key activities such as phone usage, looking away, and hand gestures. The Object and Pose Detection Submodule employs a YOLO-based detection model for rapid and accurate identification of human keypoints and phones. The Behavior Classification Submodule then interprets these detections using rule-based temporal smoothing to determine suspicious actions based on movement patterns and head orientation. Together, these submodules ensure efficient, accurate, and interpretable detection results that mimic realistic classroom conditions while maintaining low latency.

***Recording and Monitoring Module***

This module manages all aspects of live video capture, monitoring, and recording synchronization. It continuously streams camera input at high frame rates while maintaining a separate recording pipeline for archival and review purposes. The Camera Management Submodule handles device initialization, frame acquisition, and multi-camera coordination, ensuring consistent video feed quality. The Recording Submodule captures both raw and processed streams, embedding detection overlays where applicable. Meanwhile, the Monitoring Submodule ensures real-time synchronization between live viewing and detection outputs, preserving smooth video playback without interruptions or frame loss.

***Analytics and Reporting Module***

This module transforms detection outputs into structured analytical insights for post-session review. The Data Aggregation Submodule compiles event logs,

timestamps, and classification outputs into a cohesive session record. The Behavioral Analysis Submodule processes these records to identify trends, frequencies, and anomalies in student activity. Finally, the Report Generation Submodule produces comprehensive reports in multiple formats, such as graphical summaries, charts, and tabular analytics. These reports provide instructors with actionable insights regarding classroom behavior and individual student activity, aiding in transparency and evaluation.

***Video Processing Module***

This module oversees all operations related to video handling, including both live and pre-recorded footage. The Video Upload Submodule allows users to import existing recordings for analysis, during which the system performs frame-by-frame detection and overlays bounding boxes corresponding to identified behaviors. The Processing and Annotation Submodule manages the automated marking of events and gestures, storing results for later review. In addition, the Optimization Submodule ensures efficient encoding and frame extraction, maintaining compatibility across playback systems while reducing computational overhead.

***Database Management Module***

This Module serves as the central repository for all system data, including session recordings, event logs, and analytic results. The Data Storage Submodule ensures secure and organized management of structured data, supporting efficient indexing for rapid retrieval. The Data Integrity Submodule implements validation

and backup mechanisms to prevent corruption or loss, while the *Query* Optimization Submodule enhances data access performance during real-time analytics. This module underpins the entire system's reliability by providing consistent, persistent data storage across all operations.

***Security and Access Control Module***

This module upholds the integrity of the CheatGPT system by preventing unauthorized modifications and ensuring responsible data handling. This module currently integrates a Password Verification Submodule, which introduces a password mechanism required for the deletion of recorded videos. Through this feature, only authorized users are permitted to remove stored recordings, minimizing the risk of accidental or unauthorized data loss. Although simplified in scope, this implementation establishes a foundational security measure upon Which more advanced authentication and access control features can be built into future system iterations?

***User Interface Module***

This module provides a responsive and user-friendly platform through which users interact with the system. The Dashboard Interface Submodule offers live visualization of camera feeds, real-time detection overlays, and system alerts. The Interactive Control Submodule enables users to start or stop sessions, upload videos for analysis, and view or delete stored recordings under authorized access. Additionally, the Analytics Visualization Submodule presents intuitive charts and

behavior summaries derived from the reporting engine, ensuring clarity and accessibility of information across different devices and screen sizes.

All modules within the system operate in concert through a unified event-handling and data exchange framework. The integration layer manages real-time communication between the detection, recording, and analytics modules, ensuring data consistency and operational stability. It employs synchronized message queues and asynchronous processing to minimize latency and resource contention. Moreover, system-wide logging and diagnostic mechanisms provide transparency for debugging and performance monitoring. This modular integration ensures scalability, maintainability, and robustness across all components of the CheatGPT system.

***Testing Procedures***

This section details the methodologies applied in evaluating the Cheat GPT Gesture Processing Tool, ensuring each core module of the system meets stringent standards for reliability and user satisfaction. The procedures span three phases: unit testing, integration testing, and user acceptance testing, each designed to meticulously validate system functionality and performance in a classroom environment.

Unit testing was conducted to verify the proper operation of individual components such as student monitoring, gesture and device detection, event logging, real-time notifications, and secure data handling. This phase ensured

each module met its technical requirements before integration into a cohesive whole.

Integration testing followed, focusing on how the system's discrete modules interact and synchronize. Special attention was given to the seamless flow of data from detection to notification, the accuracy of event logs, and the stability of the user dashboard. This stage confirmed that the combined modules functioned smoothly, supporting real-time monitoring and decision-making for proctors.

For user acceptance testing, faculty members from the College of Computer Studies at City of Malabon University served as the primary evaluators. Using a structured rubric aligned with ISO/IEC 25010 software quality standards, they assessed each aspect of the system on a three-point scale: 3 ("Satisfactory/No Revision"), 2 ("Needs Improvement/Minor Revision"), and 1 ("Requires Redesign/Major Revision"). The tested criteria included system functionality, performance efficiency, detection accuracy, user interface clarity, reliability, usability, security and privacy, hardware compatibility, maintainability, and overall software quality.

The results across all participants consistently indicated high satisfaction, with all scores falling within "Satisfactory" and "Minor Revision" ranges. No module received a score of "1," confirming the robustness of the current implementation and the absence of critical faults that would necessitate major redesign. Most suggestions for improvement pertained to expanding gesture recognition, adding

security confirmations for data-sensitive actions, and refining instructional resources for new users.

Throughout this testing process, any identified issues were addressed promptly by adding password protection for accidental deletion, enhancing gesture coverage, and updating user guides based on evaluators' recommendations. The debugging and improvement cycle reinforced system reliability and confirmed compliance with quality benchmarks.

In summary, the comprehensive testing methodology, spanning individual module validation to user-centered acceptance, showed that the Cheat GPT system fulfills its intended purpose and maintains operational integrity under realistic classroom conditions. The evaluators' feedback, focused on refinement rather than overhaul, highlights the system's readiness for deployment and future enhancement within academic settings.

***Test Results and Debugging***

The Cheat GPT (Gesture Processing Tool) underwent a thorough, multi-stage evaluation to ensure it met necessary standards for functionality, stability, and user acceptance. The methodology included three core phases: Unit Testing, Integration Testing, and User Acceptance Testing (UAT). Each phase scrutinized a distinct aspect of system validation, combining technical assessment with practical usability.

Unit Testing focused on verifying each software module, such as detection, logging, reporting, and notifications, to confirm their correctness and internal

reliability. Integration Testing examined how these components operated together, emphasizing the seamless flow of data between monitoring, gesture recognition, and user interface elements. User Acceptance Testing brought in actual classroom professors, who interacted directly with the fully developed system and rated its performance through a structured, criteria-based questionnaire.

Evaluators used a standardized form based on a 3-point scale:

- 3 – Satisfactory / No Revision
- 2 – Needs Improvement / Minor Revision
- 1 – Requires Redesign / Major Revision

Feedback was collected for these key criteria:

- Functionality of all core modules
- Real-time performance
- Detection accuracy
- Dashboard clarity and user notifications
- Reliability (crash resistance and data stability)
- Usability (ease of use and training requirements)
- Security and privacy
- Device compatibility
- Maintainability of code and structure
- Compliance with ISO/IEC 25010 software standards

Across all evaluators, no criteria received a score of 1 (Major Revision), confirming the absence of critical flaws. Most ratings fell in the range of “3” (No Revision) and “2” (Minor Revision), highlighting consensus on the system’s operational reliability and user-friendliness. Specific remarks praised low delay and stable detection, but suggested improvements for the gesture detection scope and added security for sensitive operations like data deletion.

Debugging actions were taken to address user feedback, including:

- Adding password confirmation prompts for accidental deletions
- Expanding system-recognized gestures
- Enhancing help resources and user tutorials

The evaluation confirms that the Cheat GPT system is reliable and effective. System testers found it easy to use and stable, with only minor suggestions for improvement. Overall, the system is well-prepared to support exam proctors and help improve academic integrity. With ongoing updates and user feedback, the tool can improve further and serve different exam environments effectively.

**Interpretation of user feedback or testing outcomes**

The system's effectiveness and acceptance were systematically assessed through a Likert-scale survey administered to fifteen system testers selected via purposive sampling, representing 100% of the College of Computer Studies faculty. Each participant evaluated statements related to usability, performance efficiency, reliability, usefulness, and ethical considerations using a four-point scale, where ‘1’ denoted strong disagreement and ‘4’ indicated strong agreement.

**Table 2**

*Profile of respondents based on profession*

| Profession | Frequency | Percentage |
|---|---|---|
| CCS Faculty | 15 | 100% |
| **Total** | **15** | **100%** |

The study employed purposive sampling, a non-probability method where participants are deliberately chosen based on characteristics relevant to the research objectives. This sampling strategy ensures that individuals contributing to the study possess the specific expertise or role necessary to provide meaningful insights related to the system being evaluated.

In this study, all respondents were faculty members from the College of Computer Studies (CCS).

**Table 3**

*Overall and Section-Wise Survey Results*

| Section | Weighted Mean | Interpretation |
|---|---|---|
| Performance Efficiency | 3.822 | Strongly Agree |
| Usefulness | 3.800 | Strongly Agree |
| Reliability | 3.689 | Strongly Agree |
| Usability | 3.683 | Strongly Agree |
| Ethical Considerations | 3.644 | Strongly Agree |

| | | |
|---|---|---|
| **Overall** | **3.725** | **Strongly Agree** |

The overall feedback shows a positive response to the system. The weighted mean of 3.725 indicates that the main features and qualities meet or exceed what users expected, as shown in Table 3.

**Figure 29**

*Survey Section Weighted Means*

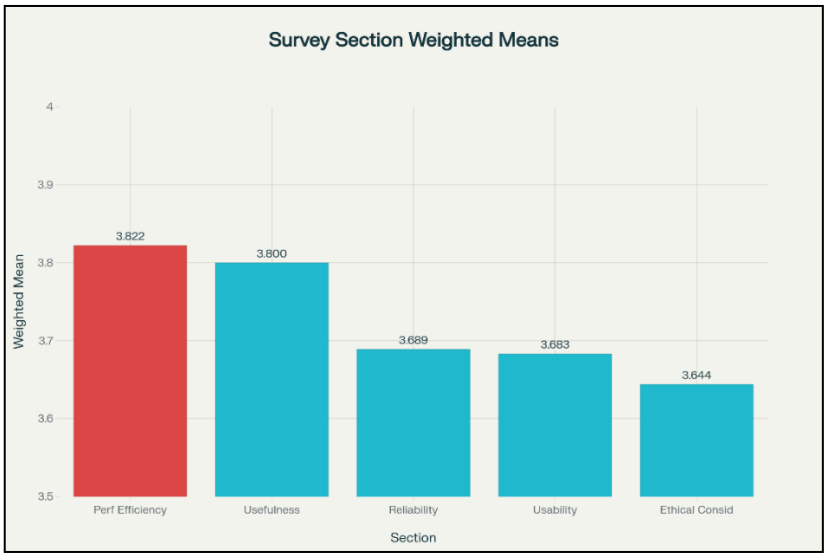



Figure 29 is a vertical bar chart that visually represents the weighted mean scores for each survey section. Performance Efficiency was most highly rated, especially the system's ability to deliver prompt, real-time alerts without lag. Usefulness and Reliability also received strong marks, reflecting the system's contribution to proctoring and stable operation. Usability remained positive, with minor suggestions for more comprehensive guides and improved training materials. Ethical Considerations, though lowest, were still firmly positive,

indicating users felt privacy and fairness were respected but saw room for even greater transparency.

**Table 4**

*Detailed Statement Analysis*

| Section | Statement Number | Weighted Mean |
|---|---|---|
| Performance Efficiency | 1 | 3.800 |
| Performance Efficiency | 2 | 3.667 |
| Performance Efficiency | 3 | 4.000 |
| Usefulness | 1 | 3.800 |
| Usefulness | 2 | 3.867 |
| Usefulness | 3 | 3.733 |
| Reliability | 1 | 3.800 |
| Reliability | 2 | 3.733 |
| Reliability | 3 | 3.533 |
| Usability | 1 | 3.733 |
| Usability | 2 | 3.667 |
| Usability | 3 | 3.667 |
| Usability | 4 | 3.667 |
| Ethical Considerations | 1 | 3.533 |
| Ethical Considerations | 2 | 3.667 |
| Ethical Considerations | 3 | 3.733 |

As seen in Table 4, system testers rated every aspect between “Agree” and “Strongly Agree.” Notably, no section or statement dropped below 3.5, confirming that users recognize the system’s solid performance, reliability, and value. The perfect score in Performance Efficiency (Statement 3) signals particular strength in the smoothness and responsiveness of real-time operation. Ethical Considerations, while still scoring very well, indicate the most growth potential, particularly in areas such as privacy transparency, user consent, and fairness assurances.

***Performance Efficiency***

This section received the highest average score (3.822), with Statement 3 earning a perfect mean of 4.000. Respondents unanimously confirmed that the system operates in real time without noticeable delay, reflecting robust strength in processing speed and alert responsiveness.

***Usefulness***

Scoring 3.800, usefulness is the second most praised dimension. Both the system’s contribution to proctoring and its capacity to provide timely, relevant information to users were consistently acknowledged. The slightly higher score for statement 2 (3.867) highlights particular appreciation for practical benefits in reducing academic dishonesty.

***Reliability***

With a weighted mean of 3.689, most evaluators agreed that the system is dependable, rarely crashes, and generally delivers accurate detection. Lower

scoring in statement 3 (3.533) compared to the other reliability factors suggests minor improvements could be made to further reduce false alarms and detection errors.

***Usability***

Usability achieved a mean of 3.683, indicating that the interface and navigation are straightforward. Evaluators consistently agreed that system instructions were clear, though there is some room to improve training materials and enhance the intuitiveness for first-time users.

***Ethical Considerations***

Although this section had the lowest mean (3.644), the score still falls solidly within the “Strongly Agree” range. Scores suggest the majority felt student privacy is respected and that the monitoring process is fair and non-invasive. Statement 1 (3.533) was the lowest individual score in the entire survey, signaling possible areas to clarify system privacy protocols and strengthen ethical transparency.

These findings show that the system delivers on its core promises, yet sustained attention to privacy education and continuous user consultation will further strengthen trust and ethical alignment.

# Chapter 4

## SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents a summary of the key findings, conclusions drawn from the research, and recommendations for future action based on the study's results. It synthesizes the information presented in the preceding chapters to provide a clear and concise overview of the research journey and its outcomes. The chapter is structured to provide a comprehensive wrap-up of the study, highlighting the main contributions and suggesting practical implications.

### Summary of Findings

1. The summary of findings from this capstone study shows important points about how the Cheat GPT system was built and how well it works in stopping cheating during classroom exams. The system combines different parts, like recognizing student gestures, detecting mobile devices, and using AI to monitor exams, to create one tool that watches for cheating in real time. This setup helps to cover what human proctors often miss, especially when they need to watch many students at once. Tests showed the AI models work well under normal classroom settings, catching cheating movements and unauthorized device use with few mistakes.
2. The system was also checked using a recognized quality standard called the ISOIEC 25010 model. This review showed the system is easy to use, works dependably, and runs efficiently. Both teachers and students said the

system is easy to interact with and does not disturb the exam flow. This made the system an acceptable tool that helps lower the amount of work for proctors while keeping the exam fair. The researchers also designed the system with privacy and consent in mind, which helped students and teachers trust it because it respects their rights.

3. Some challenges were found, like the system performing less well when classroom lighting is poor or when cameras are not positioned correctly. These issues show that using the system well means following the setup instructions carefully. Overall, this study proves that Cheat GPT is a useful and practical tool that solves the problem of not enough human proctoring during exams. It helps make exams fair and trusted. Its success gives a strong base for improving the system further and for schools to use it widely to keep standards high in academic honesty.

**Conclusions**

Based on the findings of the study, the following conclusions were drawn: The system's ability to function together across modules directly responds to the problems identified with manual proctoring lapses, proving that the Cheat GPT can assist proctors by providing continuous and objective monitoring. The high accuracy and favorable user satisfaction highlight the system's success in improving academic honesty by detecting and notifying suspicious behavior in real time, thereby upholding the integrity of classroom exams. The selected machine

learning models are suitable for the educational context, balancing detection performance and operational efficiency, making Cheat GPT a practical tool for institutions like City of Malabon University. Ethical considerations incorporated into the system's design, including data privacy safeguards and informed consent, align with users' acceptance and ensure the monitoring tool is perceived as fair and non-invasive. While the system is effective, its reliance on environmental conditions and hardware capabilities indicates that adherence to implementation guidelines is crucial for sustained performance and accuracy.

**Recommendations**

Considering the conclusions drawn, the following recommendations are proposed:

1. Institutional adoption of the Cheat GPT system is recommended to enhance the supervision of classroom exams, particularly in large classes where individual proctor observation is challenging.

2. Future improvements should focus on expanding gesture recognition capabilities and integrating multi-camera setups to overcome limitations related to occlusion and camera positioning.

3. Regular training and orientation sessions for faculty and students should be conducted to ensure a clear understanding of the system's ethical protocols and operational guidelines, reinforcing trust and transparency.

4. Further research is encouraged to optimize the system's performance in diverse classroom environments, including integration with hybrid and online examination settings to adapt to evolving educational modes.

5. Continuous monitoring and refinement should be established to address false positive rates and incorporate user feedback, ensuring the system remains reliable, effective, and aligned with academic integrity goals.

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